

Final Project Report

Multivariate Forecasting and Predictability Analysis of Cryptocurrency Market Dynamics

GitHub Repository: <https://github.com/knochez/crypto-multivar-forecast-p27gA>

2026 Capstone Group 27 A

Carlton Hogan | Sambhav Gurung | Kevin Nochez | Huijoon Jang | Alexandros Mullane | Alvin Ji
Shan Tiong

Table of Contents

Table of Contents	2
1.0 Executive Summary	3
1.1 Problem Being Solved	3
1.2 Main Findings	3
1.3 Main Recommendations	4
2.0 Business Problem	4
2.1 Problem Context	4
2.2 Project Objective	5
2.3 Literature Review	6
3.0 Data Exploration	8
3.1 Data Source	8
3.2 Data Description	9
3.3 Data Preprocessing & Cleaning	10
3.4 Exploratory Data Analysis	14
4.0 Modelling	18
4.1 Modelling Framework (Models, Four Targets)	18
4.2 Feature Selection & Lookback Windows	19
4.3 Modelling Challenges & Solutions	19
4.4 Model Evaluation & Quality	21
5.0 Findings	31
5.1 Volatility - The Most Predictable Target	31
5.2 Price - Trend Tracking, Not Direction	32
5.3 Return - Near White Noise	33
5.4 Volume - Spike-Driven, Hardest Target	34
5.5 Cross-Model Comparison & Bonus ChatGPT-Hybrid Benchmark	35
6.0 Recommendations	39
6.1 Add External Data Sources (Sentiment, On-Chain, Macro)	39
6.2 Match Model Architecture to Target Variable	39
6.3 Use the Right Metric per Target (RMSE + MDA)	40
6.4 Integrate into the DSI Research Workflow	41
7.0 Deployment, Future Work and Reflection	41
7.1 Deployment Considerations & Constraints	41
7.2 Integration into Business Information Systems	42
7.3 Future Work & Research Extensions	43
7.4 Lessons Learned	44
7.5 Group Reflection & Closing	44
8.0 Difficulties Encountered	45
8.1 Modelling & Architecture Difficulties	45
8.2 Data, Feature & Non-Stationarity Issues	46
8.3 Workload, Timeline & Coordination	46
9.0 References	47
Appendix	50

1.0 Executive Summary

1.1 Problem Being Solved

This project was completed for the Data Science Institute to test how far machine learning can forecast a noisy and non-stationary financial time series using historical market data. Cryptocurrency was used as the testbed because Bitcoin and Ethereum contain many of the problems that make financial forecasting difficult, including sharp price movements, regime changes, volatility clustering and sudden shocks.

The project does not ask whether cryptocurrency is simply predictable or unpredictable. That question is too broad. Instead, it separates the market into four target variables, being close price, log return, 30-day rolling volatility and trading volume. This matters because each target behaves differently. Prices can look predictable because today's price is usually close to yesterday's price. Returns are harder because daily patterns are quickly traded away. Volatility is more structured because high-volatility periods tend to persist. Volume is harder again because large spikes are often caused by news, sentiment and liquidation events that are not visible in historical price data.

1.2 Main Findings

The main finding is that predictability depends more on the target variable than the model itself. No single model worked best across everything. Instead, each target revealed a different level of signal.

- Volatility was the most predictable target because volatility clustering gives the models a real pattern to learn. The models tracked the broad level of volatility well, although daily direction still stayed close to random.
- Price was trackable but misleading. The models followed the broad price trend because price is highly autocorrelated, but this did not mean they could predict direction. This was clear when every model missed the October 2025 Bitcoin spike to around \$125,000.
- Log return was close to white noise. Most models predicted values near zero, and Mean Directional Accuracy did not move meaningfully above 50 percent. This suggests the models were not finding a reliable daily return signal.

- Trading volume was difficult because the largest spikes were driven by external events. Some LSTM volume models also became unstable at longer lookbacks, producing negative predictions or large overestimates.

Overall, the results show that magnitude is easier to estimate than direction. RMSE, MAE and MAPE can make a model look useful when it is only tracking the level of a series. MDA is needed because it shows whether the model has real directional value. Across the project, MDA staying near 50 percent was the clearest sign that low error does not always mean useful forecasting skill.

1.3 Main Recommendations

The project should be treated as a research pipeline rather than a trading system. The models help show where signal exists, but they do not yet support live trading because daily directional accuracy remains weak.

The next stage should focus on three improvements.

- Add external data sources such as news sentiment, on-chain activity, liquidation data, exchange flows, funding rates and macro indicators.
- Tune models separately for each target instead of using one model setup across price, return, volatility and volume.
- Keep RMSE and MDA together in the evaluation framework so that both forecast size and directional skill are assessed.

The most useful path forward is to build on the volatility results, where historical data contains the strongest signal, while improving return and volume models with better external features.

2.0 Business Problem

2.1 Problem Context

The Data Science Institute applies data science, machine learning and analytics to complex datasets. For this project, the Institute engaged our team to investigate how predictable cryptocurrency market dynamics are using Bitcoin and Ethereum data.

The client problem is not to build a trading system. A trading system would need portfolio rules, transaction costs, execution testing and risk controls. This project comes before that stage. It asks whether the underlying market variables contain enough signal to justify forecasting work in the first place.

Cryptocurrency is a useful testbed because it is hard to forecast. Bitcoin and Ethereum trade continuously, react quickly to new information and move through different regimes. A model trained on one period may not work well in another because the market structure changes.

There are three main issues that make the problem difficult:

- Market efficiency limits return prediction. If a simple daily return pattern exists, algorithmic traders can trade against it until the opportunity disappears.
- Non-stationarity weakens model transfer. The 2021 bull market, 2022 drawdown and 2024 to 2025 recovery period behave differently, so past patterns may not hold.
- External shocks limit historical-data models. News, regulation, liquidation cascades and sentiment shifts can move price and volume, but these causes are not contained in OHLCV data.

These issues explain the logic of the project. Weak return prediction may reflect market efficiency rather than poor modelling. Stronger volatility prediction suggests that volatility contains persistent structure. Missed volume spikes suggest that external explanatory variables are needed. The project therefore maps what machine learning can and cannot extract from historical crypto data.

2.2 Project Objective

The project objective is to measure the limits of predictability in daily cryptocurrency markets by benchmarking multiple model approaches on identical inputs, identical out-of-sample test data and identical evaluation metrics.

The project compares models with different levels of complexity.

- Linear Regression acts as the transparent baseline.

- Random Forest tests whether non-linear feature relationships improve performance.
- LSTM tests whether sequence-aware memory adds value.
- GRU provides a simpler recurrent benchmark against LSTM.
- The GPT-driven hybrid benchmark tests whether newer AI-assisted methods add anything beyond trained-from-scratch models.

The four target variables were chosen because each captures a different part of market behaviour. Close price represents the market level. Log return captures daily change. The 30-day rolling volatility captures risk and instability. Trading volume captures market activity. This makes the question of predictability more precise, because a model may track price well but fail on direction, predict volatility because it persists, fail on returns because they are close to noise, and fail on volume because spikes are driven by external events.

The project also tests lookback windows of 3, 5, 8 and 10 days. This allows the analysis to test whether each target benefits from short memory or slightly longer historical context. Short windows are more responsive but more exposed to noise. Longer windows smooth the data but may lag when the market regime changes.

The final deliverables are a comparative predictability hierarchy across the four targets, documented model configuration insights, and a modular Python pipeline covering data ingestion, feature engineering, model training and evaluation. These outputs allow the DSI to see which targets are most learnable, which models are most useful for each target, and what extra data would be needed to improve future forecasting work.

2.3 Literature Review

The academic literature on cryptocurrency forecasting has developed through three broad stages. The first stage focused on conventional statistical models such as ARIMA, VAR and GARCH. These methods created useful benchmarks, especially for volatility forecasting. Catania, Grassi and Ravazzolo (2019) tested univariate and multivariate models across major cryptocurrencies and found that forecast combinations can outperform individual models. Conrad, Custovic and Ghysels (2018) extended this work through a GARCH-MIDAS framework, showing that broader market volatility can influence long-term Bitcoin volatility. These studies show that crypto markets are not completely separate from wider financial conditions. However, traditional models are limited because cryptocurrency data is noisy, non-linear and unstable. Crypto returns do

not consistently satisfy assumptions of linearity, normality or stationarity, which reduces the usefulness of purely statistical approaches (Chen et al. 2021; Khedr et al. 2021).

The second stage moved toward conventional machine learning. This shift was logical because machine learning models can capture non-linear relationships that traditional models may miss. Kristjanpoller and Minutolo (2018) found that combining neural networks with GARCH methods improved Bitcoin volatility forecasting. Nakano, Takahashi and Takahashi (2018) showed that an MLP model trained on technical indicators could outperform a naive buy-and-hold baseline for Bitcoin returns. McNally, Roche and Caton (2018) extended this to Bitcoin price prediction and found that recurrent neural networks could outperform ARIMA on directional classification. Ibrahim, Kashef and Corrigan (2021) also compared ARIMA, Prophet, MLP and Random Forest for cryptocurrency direction prediction. The overall lesson from this stage is that machine learning can extract patterns missed by classical models, but it can also overfit when the dataset is small, noisy or affected by regime change (Hajek, Hikkerova & Sahut 2023).

The third stage focuses on deep learning and ensemble methods. LSTM and GRU models are important because they are designed for sequential data and can retain information across time steps (Hochreiter & Schmidhuber 1997). Lahmiri and Bekiros (2019) found that LSTM can outperform conventional machine learning methods in Bitcoin forecasting. However, this does not mean LSTM is always the best model. Zhang et al. (2021) found that GRU can outperform LSTM in some cryptocurrency settings because it is simpler and less prone to overfitting. Murray et al. (2023) also found that recurrent architectures remain competitive with newer deep learning models. This supports the use of LSTM in our project, while also showing why its performance must be tested rather than assumed (Khedr et al. 2021; Zhang, Cai & Wen 2024).

Ensemble methods provide another important comparison point. Sun, Liu and Sima (2020) found that LightGBM can outperform Random Forest for cryptocurrency direction forecasting, suggesting that boosted models may be better suited to some crypto prediction tasks. Sebastião and Godinho (2021) showed that machine learning performance changes across market conditions, while Manchanda and Aggarwal (2021) found that simpler ensemble methods can sometimes compete with deep learning models. Most recently, Bouteska et al. (2024) compared ensemble and deep learning methods across several cryptocurrencies and concluded that no single architecture dominates across all assets and settings. This finding is directly relevant to our project because it supports our decision to compare multiple model families rather than search for one universal best model.

The literature also highlights why evaluation design matters. Many studies rely heavily on RMSE or MAPE, which measure the size of the error but do not show whether the model can predict direction. Hyndman and Koehler (2006) also note that percentage-based error measures can become unstable when actual values are close to zero. This is especially important for return forecasting. A model can achieve a low RMSE on price by following the level of the series, while still failing to predict whether the next movement will be up or down. For this reason, our project uses RMSE, MAE, MAPE and Mean Directional Accuracy together. This allows us to separate level tracking from genuine directional forecasting skill.

Overall, the literature supports the structure of this project. Volatility should be more learnable because it tends to persist. Price can be tracked in level because it is autocorrelated. Returns should be difficult because exploitable short-term patterns are quickly traded away. Volume should be difficult because the largest movements are often caused by external events missing from historical OHLCV data. The literature also leaves space for our GPT-hybrid benchmark, because AI-assisted forecasting methods are still not well established against trained-from-scratch machine learning models in this specific setting. Our project therefore builds on the existing literature by comparing several architectures across four targets, four lookback windows and a 2025 out-of-sample test set.

3.0 Data Exploration

3.1 Data Source

The dataset used in this project was obtained from Binance's public market data API. This source was selected because it provides historical cryptocurrency trading data for our targeted cryptocurrencies, Bitcoin and Ethereum. The dataset covers the period from 2021-01-01 to 2025-12-31 and is recorded at daily intervals.

Historical market data was collected using a Python script that accessed the API to retrieve daily candlestick (kline) data for the BTCUSDT and ETHUSDT trading pairs. Specifically, the script automatically downloaded the data for the specified time range and extracted key trading variables from the API response. The raw data was then

processed and organised into an XLSX dataset with separate sheets for each cryptocurrency.

Daily data is suitable for this project because it provides sufficient historical observations for forecasting price, return, volatility and trading volume. It also reduces the impact of excessive noise that may appear in intraday cryptocurrency data. Additionally, the selected time from 2021 to 2025 covers different market conditions including growth periods, market declines, and high-volatility periods. As a result, the dataset is appropriate for analysing cryptocurrency volatility and predictability.

3.2 Data Description

The raw dataset consisted of 10 original market variables and 1,826 observations for both BTCUSDT and ETHUSDT. The original variables included date, price-related variables, and trading activity variables.

After feature engineering, the number of variables increased from 10 to 30 per asset by adding 20 derived variables. These engineered variables were created to capture return, volatility, trend, momentum, RSI, and volume-based patterns. Table 1 summarises the main variable groups used in the final dataset.

Category	Variables	Purpose
Time variable	date	Identifies the trading date of each observation
Price variables	open, high, low, close	Describe daily price movement and price level
Trading activity variables	volume, quote_asset_volume, number_of_trades, taker_buy_base_asset_volume, taker_buy_quote_asset_volume	Describe market participation and trading activity
Return variables	daily_return, log_return	Measure daily price change
Volatility variables	volatility_7, volatility_14, volatility_30	Capture market instability over different rolling windows
Trend indicators	MA_7, MA_14, MA_30, EMA_7, EMA_14, EMA_30	Capture short-term and medium-term price trends
Momentum indicators	momentum_7, momentum_14, momentum_30	Measure the strength and direction of recent price movement

RSI indicators	RSI_7, RSI_14, RSI_30	Reflect overbought and oversold market conditions
Volume indicators	volume_MA_7, volume_MA_14, volume_MA_30	Capture changes in trading volume over time

Table 1: Summary of Variables in the Final Dataset

These variables provided a broader representation of cryptocurrency market behaviour and supported the modelling of price, return, volatility, and trading volume. A full data dictionary with detailed descriptions of each variable is provided in Appendix A.

After cleaning, the final dataset retained 1,796 observations per asset after removing rows affected by indicator-related missing values. With no remaining data quality issues, the final modelling dataset was complete and ready for forecasting analysis.

3.3 Data Preprocessing & Cleaning

```

=== BTC INFO ===
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1826 entries, 0 to 1825
Data columns (total 10 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   date                                 1826 non-null  datetime64[ns]
1   open                                1826 non-null  float64
2   high                                1826 non-null  float64
3   low                                 1826 non-null  float64
4   close                               1826 non-null  float64
5   volume                              1826 non-null  float64
6   quote_asset_volume                  1826 non-null  float64
7   number_of_trades                    1826 non-null  int64
8   taker_buy_base_asset_volume         1826 non-null  float64
9   taker_buy_quote_asset_volume        1826 non-null  float64
dtypes: datetime64[ns](1), float64(8), int64(1)

=== ETH INFO ===
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1826 entries, 0 to 1825
Data columns (total 10 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   date                                 1826 non-null  datetime64[ns]
1   open                                1826 non-null  float64
2   high                                1826 non-null  float64
3   low                                 1826 non-null  float64
4   close                               1826 non-null  float64
5   volume                              1826 non-null  float64
6   quote_asset_volume                  1826 non-null  float64
7   number_of_trades                    1826 non-null  int64
8   taker_buy_base_asset_volume         1826 non-null  float64
9   taker_buy_quote_asset_volume        1826 non-null  float64
dtypes: datetime64[ns](1), float64(8), int64(1)

```

Figure 1: BTCUSDT and ETHUSDT Dataset Information

After obtaining the BTCUSDT and ETHUSDT datasets, data quality was checked step by step before modelling. First, the overall structure of both datasets was examined, specifically the number of rows, columns, and data types. Since both datasets had the same number of rows and columns, and the data types were correctly formatted, no data type corrections were required.

```

=== BTC MISSING VALUES ===
date           0
open           0
high           0
low            0
close          0
volume         0
quote_asset_volume 0
number_of_trades 0
taker_buy_base_asset_volume 0
taker_buy_quote_asset_volume 0
dtype: int64

=== ETH MISSING VALUES ===
date           0
open           0
high           0
low            0
close          0
volume         0
quote_asset_volume 0
number_of_trades 0
taker_buy_base_asset_volume 0
taker_buy_quote_asset_volume 0
dtype: int64

=== BTC DUPLICATES ===
Duplicate rows: 0
Duplicate dates: 0

=== ETH DUPLICATES ===
Duplicate rows: 0
Duplicate dates: 0

```

Figure 2: Missing Values and Duplicates Check

Next, missing values and duplicate records were checked, especially for the date column. No missing values or duplicate dates were found in the original BTCUSDT and ETHUSDT datasets.

```

print("BTC:", btc["date"].is_monotonic_increasing)
print("ETH:", eth["date"].is_monotonic_increasing)

BTC: True
ETH: True

```

Figure 3: Date Column Chronological Order Check

The date column was also checked and confirmed to be in chronological order for both cryptocurrencies. This was a crucial step because the project used time-series data, where the order of observations affects the construction of lag-based and rolling-window features.

Feature engineering was then applied to extend the dataset beyond the original 10 market variables. The original variables mainly covered daily price levels and trading activity. Although these original variables were useful, they were not enough to fully support the forecasting of all four target variables: price, return, volatility, and trading volume. To capture return changes, volatility patterns, trend, momentum, and changes in market participation, 20 additional engineered variables were created to provide the models with more useful information for forecasting.

The engineered features were designed to capture different types of market behaviour. Return-based features measured day-to-day price changes, while rolling volatility features captured market instability over different time windows. Moving averages and

exponential moving averages represented price trends, momentum indicators captured the strength and direction of recent price movements, RSI reflected overbought and oversold conditions, and volume moving averages represented changes in market participation.

```

└─
=== BTC NEW INFO ===
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1826 entries, 0 to 1825
Data columns (total 30 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   date                                     1826 non-null   datetime64[ns]
1   open                                    1826 non-null   float64
2   high                                    1826 non-null   float64
3   low                                     1826 non-null   float64
4   close                                   1826 non-null   float64
5   volume                                  1826 non-null   float64
6   quote_asset_volume                     1826 non-null   float64
7   number_of_trades                       1826 non-null   int64
8   taker_buy_base_asset_volume            1826 non-null   float64
9   taker_buy_quote_asset_volume           1826 non-null   float64
10  daily_return                           1825 non-null   float64
11  log_return                             1825 non-null   float64
12  volatility_7                           1819 non-null   float64
13  volatility_14                          1812 non-null   float64
14  volatility_30                          1796 non-null   float64
15  MA_7                                    1820 non-null   float64
16  MA_14                                  1813 non-null   float64
17  MA_30                                  1797 non-null   float64
18  EMA_7                                    1826 non-null   float64
19  EMA_14                                  1826 non-null   float64
20  EMA_30                                  1826 non-null   float64
21  momentum_7                             1819 non-null   float64
22  momentum_14                            1812 non-null   float64
23  momentum_30                            1796 non-null   float64
24  volume_MA_7                             1820 non-null   float64
25  volume_MA_14                            1813 non-null   float64
26  volume_MA_30                            1797 non-null   float64
27  RSI_7                                    1819 non-null   float64
28  RSI_14                                  1812 non-null   float64
29  RSI_30                                  1796 non-null   float64
dtypes: datetime64[ns](1), float64(28), int64(1)

└─
=== ETH NEW INFO ===
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1826 entries, 0 to 1825
Data columns (total 30 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   date                                     1826 non-null   datetime64[ns]
1   open                                    1826 non-null   float64
2   high                                    1826 non-null   float64
3   low                                     1826 non-null   float64
4   close                                   1826 non-null   float64
5   volume                                  1826 non-null   float64
6   quote_asset_volume                     1826 non-null   float64
7   number_of_trades                       1826 non-null   int64
8   taker_buy_base_asset_volume            1826 non-null   float64
9   taker_buy_quote_asset_volume           1826 non-null   float64
10  daily_return                           1825 non-null   float64
11  log_return                             1825 non-null   float64
12  volatility_7                           1819 non-null   float64
13  volatility_14                          1812 non-null   float64
14  volatility_30                          1796 non-null   float64
15  MA_7                                    1820 non-null   float64
16  MA_14                                  1813 non-null   float64
17  MA_30                                  1797 non-null   float64
18  EMA_7                                    1826 non-null   float64
19  EMA_14                                  1826 non-null   float64
20  EMA_30                                  1826 non-null   float64
21  momentum_7                             1819 non-null   float64
22  momentum_14                            1812 non-null   float64
23  momentum_30                            1796 non-null   float64
24  volume_MA_7                             1820 non-null   float64
25  volume_MA_14                            1813 non-null   float64
26  volume_MA_30                            1797 non-null   float64
27  RSI_7                                    1819 non-null   float64
28  RSI_14                                  1812 non-null   float64
29  RSI_30                                  1796 non-null   float64
dtypes: datetime64[ns](1), float64(28), int64(1)

```

Figure 4: Dataset Information after Feature Engineering

After feature engineering, the datasets were checked again to ensure that the new features were added correctly. The newly added columns had the expected data types, and the number of variables increased from 10 to 30 for each cryptocurrency.

ETH missing values:		BTC missing values:	
date	0	date	0
open	0	open	0
high	0	high	0
low	0	low	0
close	0	close	0
volume	0	volume	0
quote_asset_volume	0	quote_asset_volume	0
number_of_trades	0	number_of_trades	0
taker_buy_base_asset_volume	0	taker_buy_base_asset_volume	0
taker_buy_quote_asset_volume	0	taker_buy_quote_asset_volume	0
daily_return	1	daily_return	1
log_return	1	log_return	1
volatility_7	7	volatility_7	7
volatility_14	14	volatility_14	14
volatility_30	30	volatility_30	30
MA_7	6	MA_7	6
MA_14	13	MA_14	13
MA_30	29	MA_30	29
EMA_7	0	EMA_7	0
EMA_14	0	EMA_14	0
EMA_30	0	EMA_30	0
momentum_7	7	momentum_7	7
momentum_14	14	momentum_14	14
momentum_30	30	momentum_30	30
volume_MA_7	6	volume_MA_7	6
volume_MA_14	13	volume_MA_14	13
volume_MA_30	29	volume_MA_30	29
RSI_7	7	RSI_7	7
RSI_14	14	RSI_14	14
RSI_30	30	RSI_30	30
dtype: int64		dtype: int64	

Figure 5: Missing Values after Feature Engineering

Some missing values appeared in the engineered variables after this step. This was expected because rolling-window features require previous observations before values can be calculated. For example, indicators based on 7, 14, or 30 days cannot produce complete values at the beginning of the dataset. Since the highest number of missing values in any column was 30, the first 30 rows were removed from both cryptocurrency datasets.

```

BTC shape before: (1826, 30)
BTC shape after: (1796, 30)

ETH shape before: (1826, 30)
ETH shape after: (1796, 30)

```

Figure 6: Dataset Shape before and after Cleaning

After removing these rows, the final cleaned datasets retained 1,796 observations for each cryptocurrency. With no remaining data quality issues, the final modelling dataset was complete and ready for forecasting analysis.

3.4 Exploratory Data Analysis

Before modelling, we conducted exploratory data analysis on the final cleaned datasets to identify the market characteristics and behaviour of BTCUSDT and ETHUSDT. The analysis mainly focused on price, log return, 30-day rolling volatility, and trading volume for each cryptocurrency. We also generated correlation heatmaps to see how the original variables and engineered features were related.

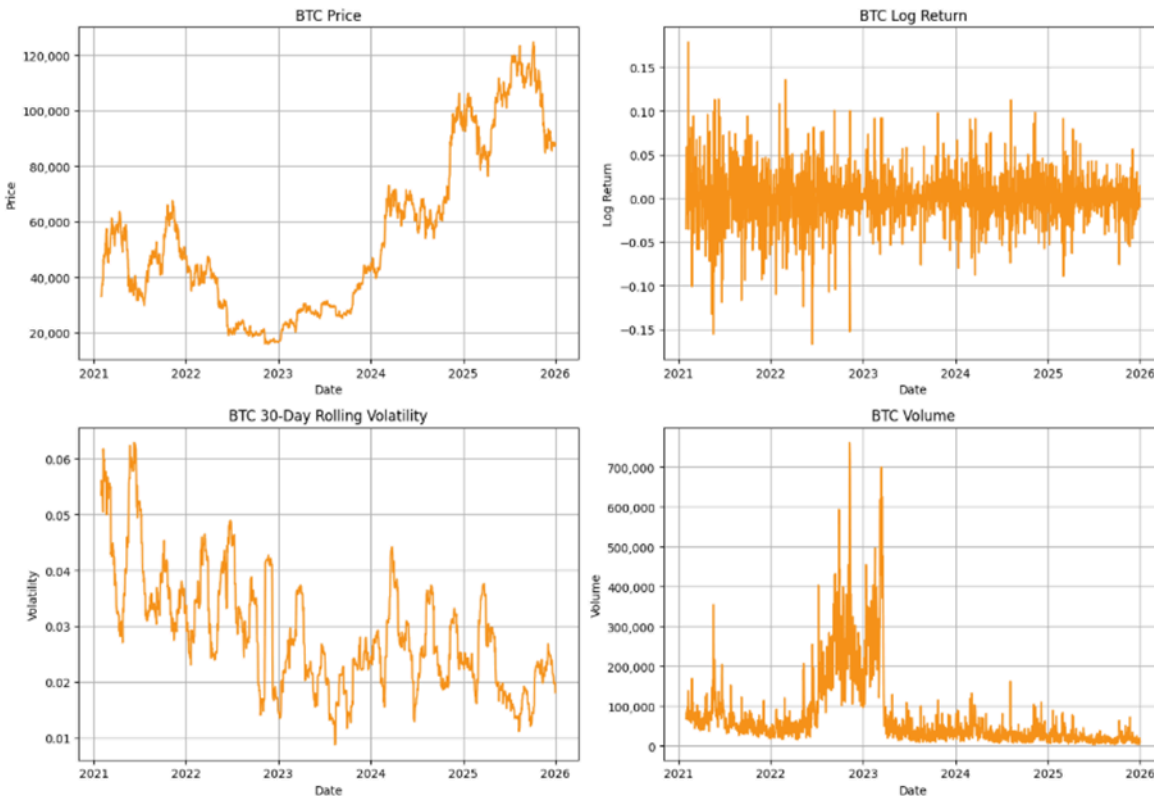


Figure 7: BTCUSDT Market Behaviour from 2021 to 2025

For Bitcoin, the price had several strong rises and falls from 2021 to 2025. The overall movement was upward, but the price was not stable across the whole period. From this, we can say that Bitcoin had dynamic price behaviour and was strongly influenced by changing market conditions over time.

Besides, we found that the log return plot was centred around zero, but there were many sudden spikes in both positive and negative directions. This means that although most daily changes were relatively small, large movements still happened quite often. Because of this, return seemed difficult to predict as it changed irregularly over time.

The 30-day rolling volatility plot also showed that volatility was not constant over time. Instead, high-volatility periods tended to continue for some time before decreasing again. This pattern is important for our modelling because it shows volatility clustering,

where volatility has some persistence over time. Compared with log return, volatility appeared to have a clearer time-series pattern.

Lastly, the trading volume plot showed several large spikes across the period rather than a stable pattern. This indicates that market activity was uneven and often increased during major market events or strong price movements.

Overall, we can conclude that Bitcoin was a highly volatile asset with changing price, return, volatility, and trading activity over time.

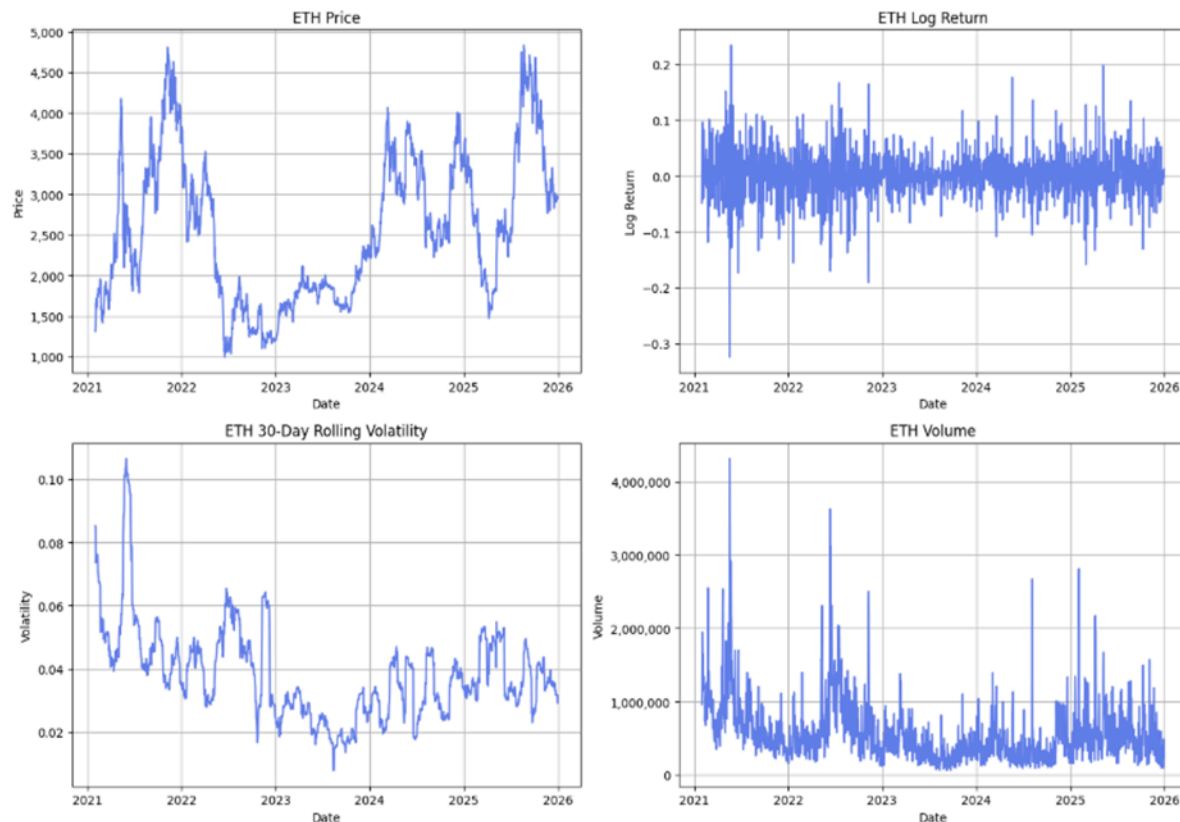


Figure 8: ETHUSDT Market Behaviour from 2021 to 2025

For Ethereum, we found that the price appeared to be more volatile than Bitcoin. From the plot, Ethereum's price rose and fell rapidly within a wide range. From this, we can say that the Ethereum market showed stronger fluctuations.

Similar to Bitcoin, we found that the ETHUSDT log return plot was also centred around zero. However, it showed larger and more frequent extreme values. This suggests that Ethereum appeared to carry a higher level of short-term risk compared with Bitcoin.

The 30-day rolling volatility plot also showed that Ethereum was more volatile than Bitcoin. It generally had stronger volatility movements and clearer volatility clustering

across the period. This again supports that our modelling should focus more on volatility because Ethereum's volatility also had some persistence over time.

Lastly, we found that Ethereum's trading volume had frequent large spikes. This means that trading activity changed sharply over time instead of remaining stable. This suggests that Ethereum's market behaviour was highly uncertain, as it was strongly affected by changes in market activity.

To summarise, Ethereum looked more unstable and reactive than Bitcoin in terms of price, return, volatility, and trading volume.

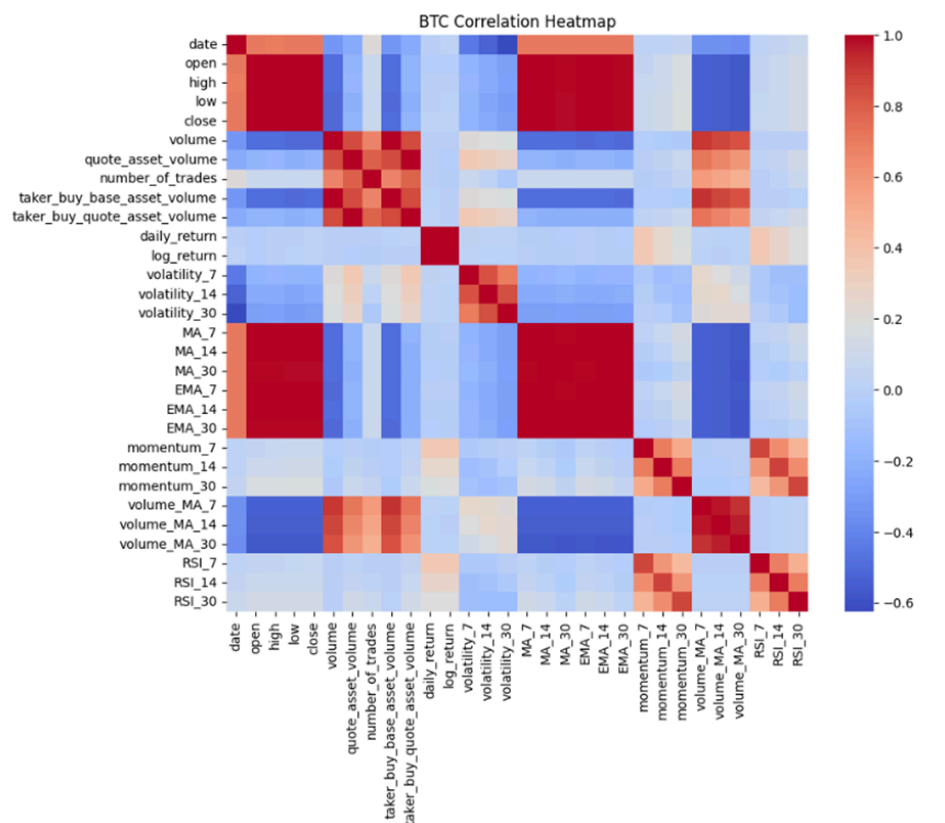


Figure 9: BTCUSDT Correlation Heatmap

For the BTCUSDT correlation heatmap, we found that the price-related variables, including open, high, low, and close, were highly positively correlated with each other. This was expected because these variables all describe similar parts of price movement. The moving averages and exponential moving averages also had strong positive relationships with the price variables, since they were calculated based on price data.

We also found that the volume-related variables were strongly correlated with each other. These included trading volume, quote asset volume, number of trades, taker buy

base asset volume, and taker buy quote asset volume. This means that trading activity variables usually moved together.

However, log return, volatility, momentum, and RSI showed weaker or more mixed correlations with price variables. This makes sense because these engineered features do not simply follow price levels. Instead, they capture other parts of market behaviour, such as return movement, market risk, momentum, and market strength. From this heatmap, we can say that the engineered features added extra information for model development.

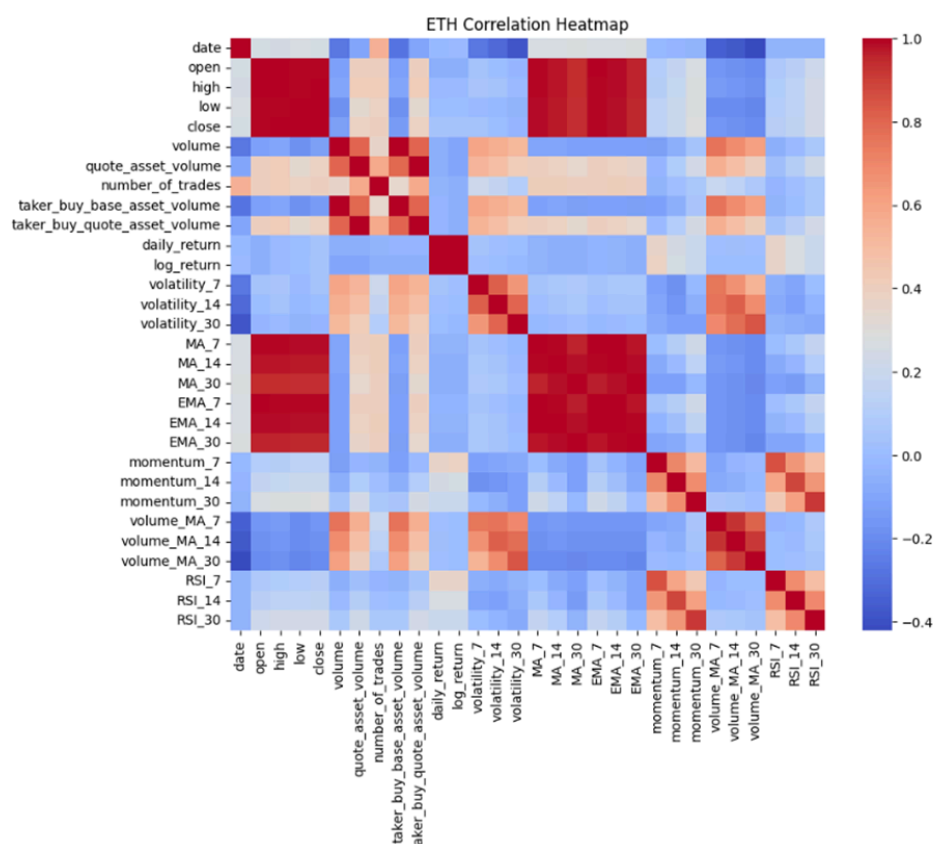


Figure 10: ETHUSDT Correlation Heatmap

For the ETHUSDT correlation heatmap, we found a similar pattern between Bitcoin and Ethereum. The price-related variables were strongly correlated with each other. MA and EMA also had strong correlations with the price variables because they reflected the overall price trend.

The volume-related variables also showed strong relationships with each other. This suggests that trading activity in the Ethereum market tended to increase or decrease together. However, compared with Bitcoin, Ethereum showed slightly more variation in some engineered features, especially volatility, momentum, and RSI. This suggests that Ethereum had more variation in market risk and short-term market behaviour.

Overall, the heatmaps showed that BTCUSDT and ETHUSDT had a similar correlation structure, but Ethereum showed slightly more variability. Together with the earlier price, return, volatility, and volume graphs, the heatmaps helped show that the engineered features captured more than just price movement. This was useful for the later modelling stage, especially when comparing how predictable price, return, volatility, and trading volume were.

4.0 Modelling

This section will describe the modelling methodology adopted for our project, it will cover the choice of model architectures, the challenges that were encountered during model development and their solutions and the evaluation framework used to assess and compare model performance. The aim of our research was not to achieve profitable prediction but rather to empirically investigate which market variables are more or less predictable under multiple model frameworks, and to compare results across four fundamentally different targets.

4.1 Modelling Framework (Models, Four Targets)

Chosen Prediction Models

- **Traditional ML Models - Linear Regression and Random Forest**

Linear Regression and Random Forest Regression were selected as the traditional machine learning models used in this report, which would serve as a basis of comparison for more complex models. Linear Regression was implemented as a baseline model due to its simplicity and clarity, while Random Forest was chosen for its more complex and non-linear relationships with the dataset.

- **Deep Learning Architectures - LSTM and GRU**

Two recurrent deep learning model architectures were employed in our research which are a Long Short-Term Memory network (LSTM) and a Gated Recurrent Unit (GRU). Both architectures sit within the branch of recurrent neural networks and are meant to deal with time series data that have temporal dependencies which makes them a good fit for financial time series where the value of a variable on any given day is influenced by recent history.

Chosen Targets

Four prediction targets were selected in this research to evaluate the different aspects of cryptocurrency behaviour. These targets include:

- **Close price:** the final trading price of a cryptocurrency at the end of each day. This target was selected because it directly reflects the market value of the asset and is one of the most commonly analysed variables in financial forecasting.
- **Log return:** the relative change in price between trading periods using a logarithmic calculation. This target was included because it focuses on price movement and direction rather than absolute price values, making it useful for analysing market behaviour.
- **The volatility:** the level of price fluctuation over a given period. It was selected to evaluate how well the models could identify changes in market stability and overall trading risk within cryptocurrency markets.
- **Trading volume:** the amount of cryptocurrency traded during a trading day. This target was included because trading activity can provide insight into market participation, liquidity, and overall investor behaviour.

4.2 Feature Selection & Lookback Windows

Feature Selection

A set of historical cryptocurrency indicators was used as input features for the predictive models. Core features included close price, log return, trading volume, and volatility, while the deep learning architectures additionally incorporated technical indicators such as RSI, EMA, momentum, and number of trades to provide more contextual market information. These features were selected to capture different aspects of cryptocurrency market behaviour, including price movement, market activity, momentum, and risk.

Lookback Windows

Multiple lookback windows of 3, 5, 8, and 10 days were also tested across all models. The lookback window determines how many previous days of historical data are used to predict future values. Testing different window sizes allowed the research to evaluate how short-term historical information influenced model performance and forecasting accuracy.

4.3 Modelling Challenges & Solutions

Traditional ML Models Problems and Solutions

- **Risk of Underfitting and Overfitting**

A common challenge in traditional machine learning models is balancing underfitting and overfitting. Simpler models such as Linear Regression may fail to capture important market patterns, while more complex models such as Random Forest can become too dependent on specific training data patterns and lose generalisability. To address this issue, multiple lookback windows of 3, 5, 8, and

10 days were tested to evaluate which historical sequence lengths produced more stable forecasting performance.

- **Sensitivity to Noise and Market Volatility**

Traditional machine learning models are also sensitive to noisy and highly volatile financial data, particularly within cryptocurrency markets where sudden market shifts are common. This made targets such as log return and trading volume difficult to predict consistently. To address this challenge, multiple target variables and evaluation metrics (RMSE, MAE, MAPE, and MDA) were used throughout the experiments to better assess model performance across different market conditions

Deep Learning Architecture Problems and Solutions

- **Dynamic Nature of Price**

Close price drifts significantly over the period being studied from approximately \$15000 to \$124000 USD for bitcoin and around \$700 to \$4000 for eth. Training a neural network on non stationary values is difficult due to the fact that the model needs to generalise to price levels in the test set that were never encountered during training. We addressed this issue by applying MinMaxScaling to all features normalising them from 0 to 1. It was important to fit the scaler on training data and applying it without refitting to the validation and test sets. This ensured that no information from future periods influenced the scaling parameters. All of the predictions are reverse transformed back to the original scale before evaluation so that all imported metrics are able to be interpreted in original units.

- **Mean Reversion Bias**

For both predictions of return and volume the models tended to predict near zero or near average values across the period which produced a flat line regardless of market movements. This challenge is well documented in the world of financial predictive forecasting caused by the means squared error loss function. It punishes large errors quadratically and becomes an incentive for the model to hedge towards the empirical mean rather than committing to extreme predictions. For volume the model is unable to anticipate spike events driven by external factors and for return the behaviour is statistically rational given the fact daily returns are centered around zero.

- **BatchNorm Single Sample Crash**

When utilising BatchNorm a batch which only contains one sample causes a runtime error because batch normalisation requires at least two samples to compute meaningful stats. This happened when the training size wasn't exactly divisible by the batch size of 32, leaving a final incomplete batch of 1. We resolved the issue by setting `drop_last = True` in the training dataloader and discarding the final incomplete batch during training. The effect this has is minimal as at most 31 samples out of over 1200 are dropped per epoch.

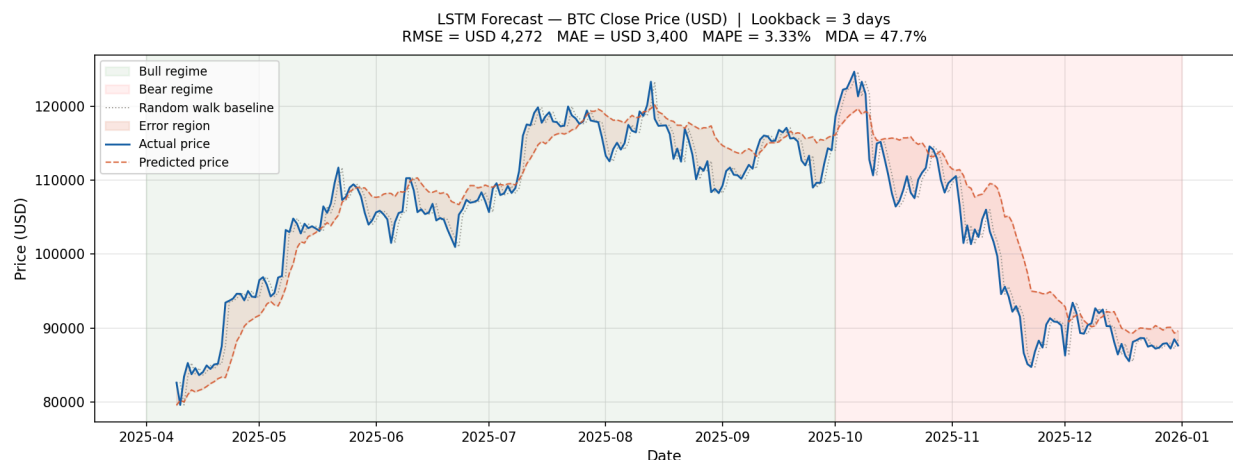
4.4 Model Evaluation & Quality

Evaluation Metrics

Four metrics were used for each model to provide a comprehensive and multidimensional assessment of the forecasting performance. Root Mean Square Error (**RMSE**), is the primary metric and measures the average size of prediction errors. Mean Absolute Error (**MAE**) measures the average absolute difference between predicted and actual values. Mean Absolute Percentage Error (**MAPE**) expresses predictions error as a percentage of the actual value. Mean Directional Accuracy (**MDA**) measures the percentage of days on which the model correctly predicts if the target variable would rise or fall, 50% baseline is equivalent to random guessing. In the results below we use the btc forecasting because it provided more consistent results.

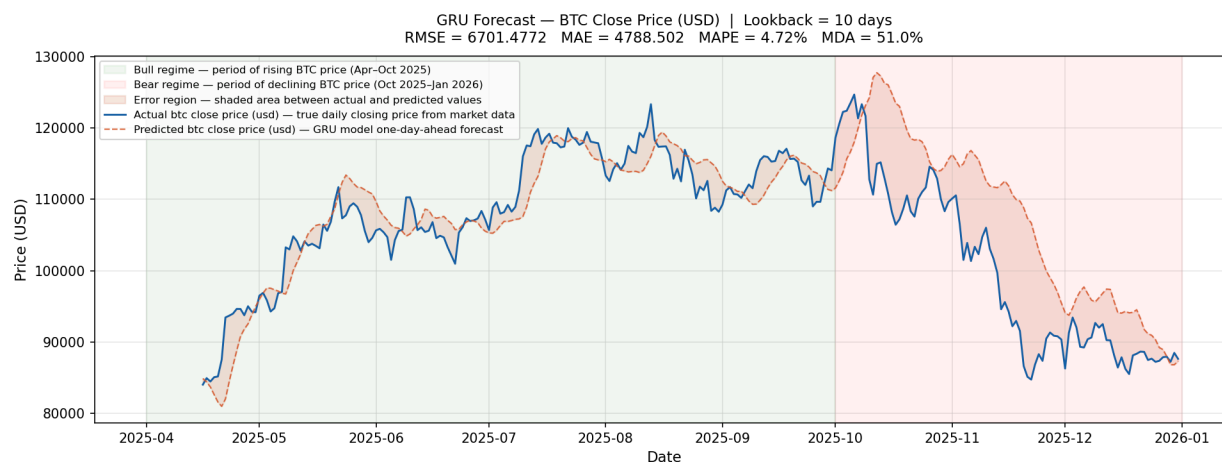
Price Forecasting

LSTM achieved its best result at lookback 3 with an RMSE of \$4272 and MAPE of 3.3% with MDA 47.7%, visually it tracked the daily trend but failed to predict daily direction.

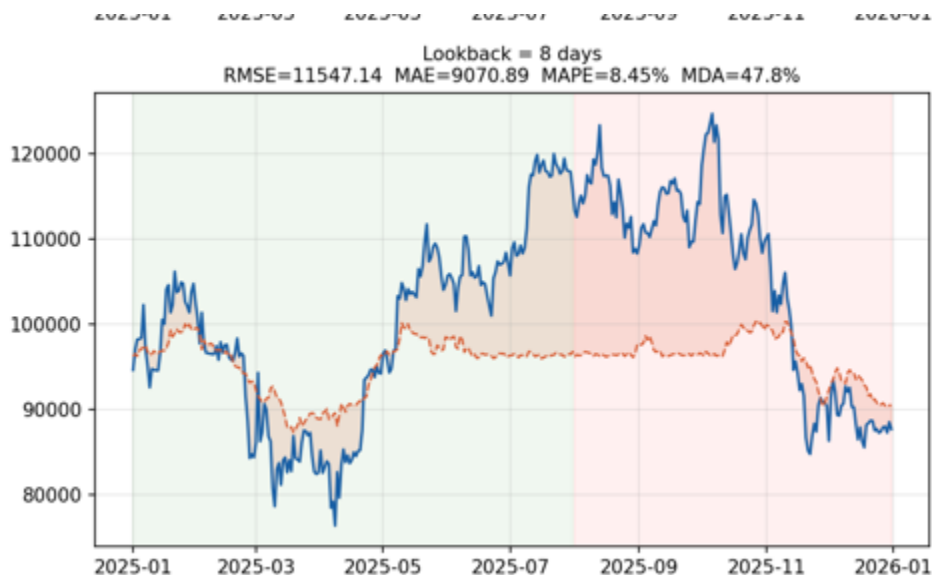


Our GRU forecasting performed best at lookback 10 with an RMSE of \$6701 and MAPE of 4.72% as well as an MDA of 51%. The magnitude accuracy was slightly weaker than

the LSTM but the GRU had better directional performance, this suggests that the LSTM's dedicated memory cell provides an advantage for non stationary price series.



Random Forest achieved its best performance at lookback 8 with RMSE of \$11547 and MAPE of 8.45% as well as MDA of 47.8%, the line it predicted was noticeably smoother than the actual price and it consistently underestimated the peaks of price, reflecting the ensemble model's tendency toward the training mean.



Linear regression produced the lowest RMSE of all models at \$2127 with MAPE 1.56% and MDA of 48.6%. This result is slightly misleading due to the fact that it exploits price autocorrelation rather than genuinely forecasting.

Linear Regres



GPT Hybrid model had an RMSE of \$2374 at lookback 3 with MAPE at 1.8% MDA at 51.2%. This performance can lead us to assume a hybrid approach adds modest but consistent directional value, meaningfully approving directional accuracy from linear regression.

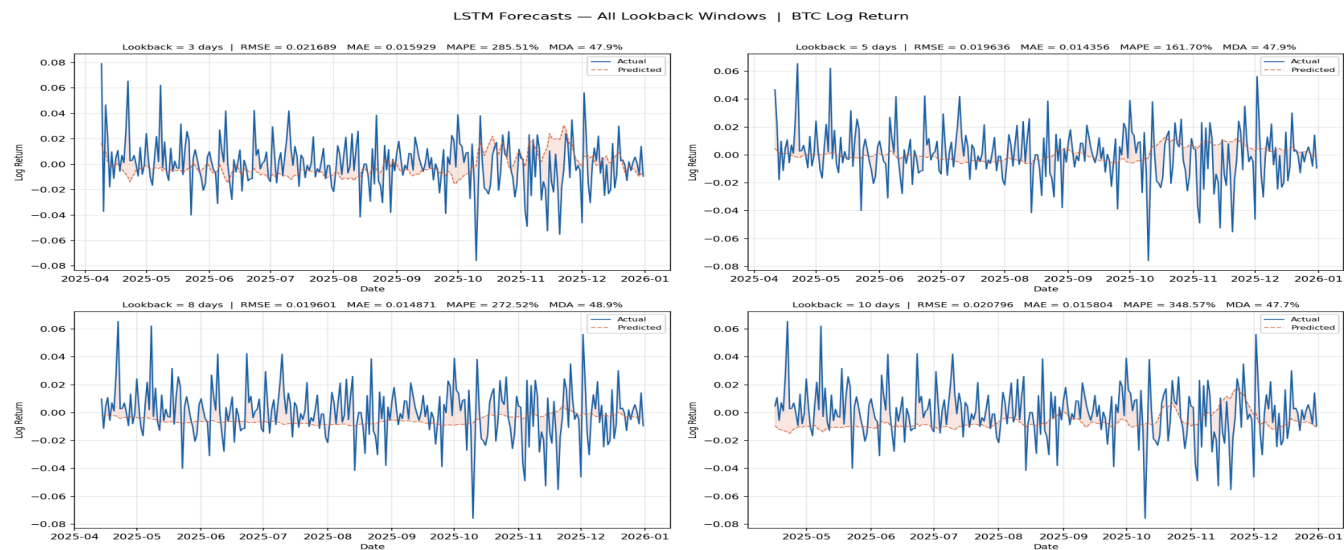
GPT-H



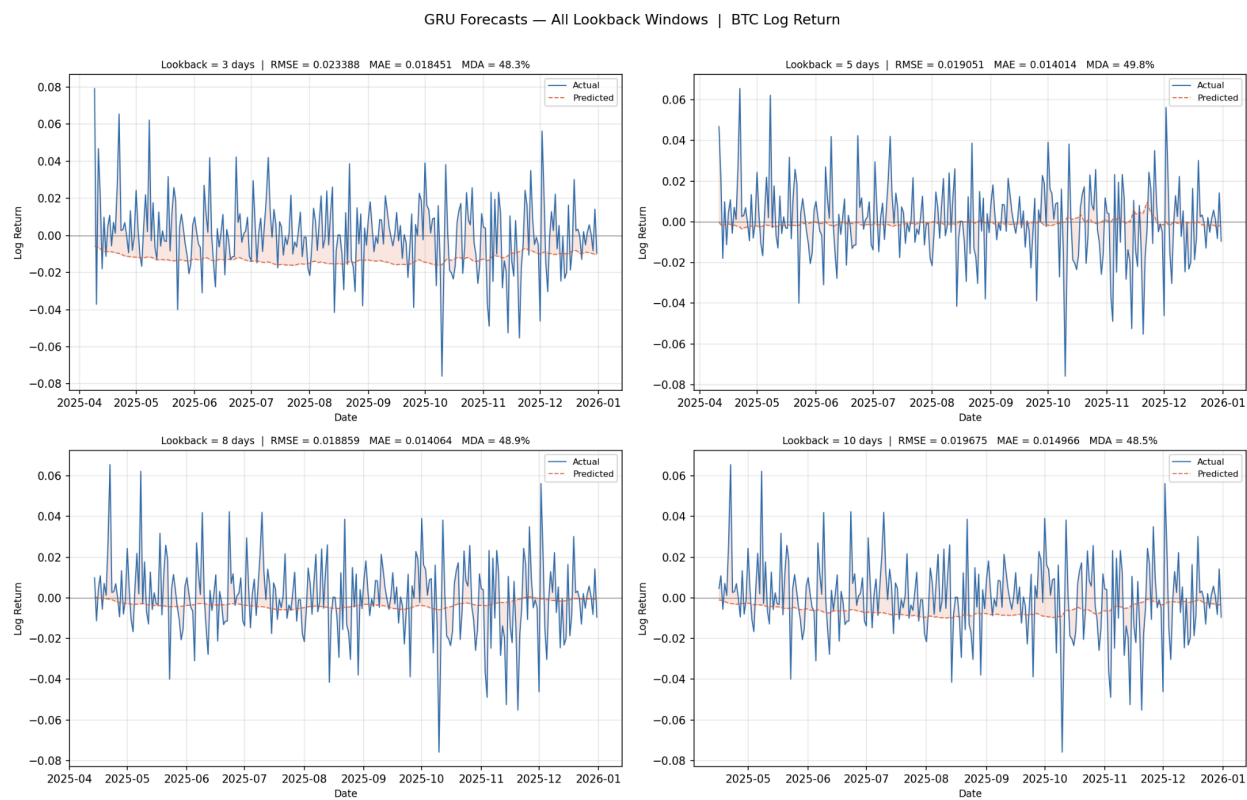
Return Forecasting

LSTM predicted flatline near zero, with RMSE of 0.01960 and MDA of 48.9%, sometimes the most expressive recurrent architecture finds no exploitable signal in

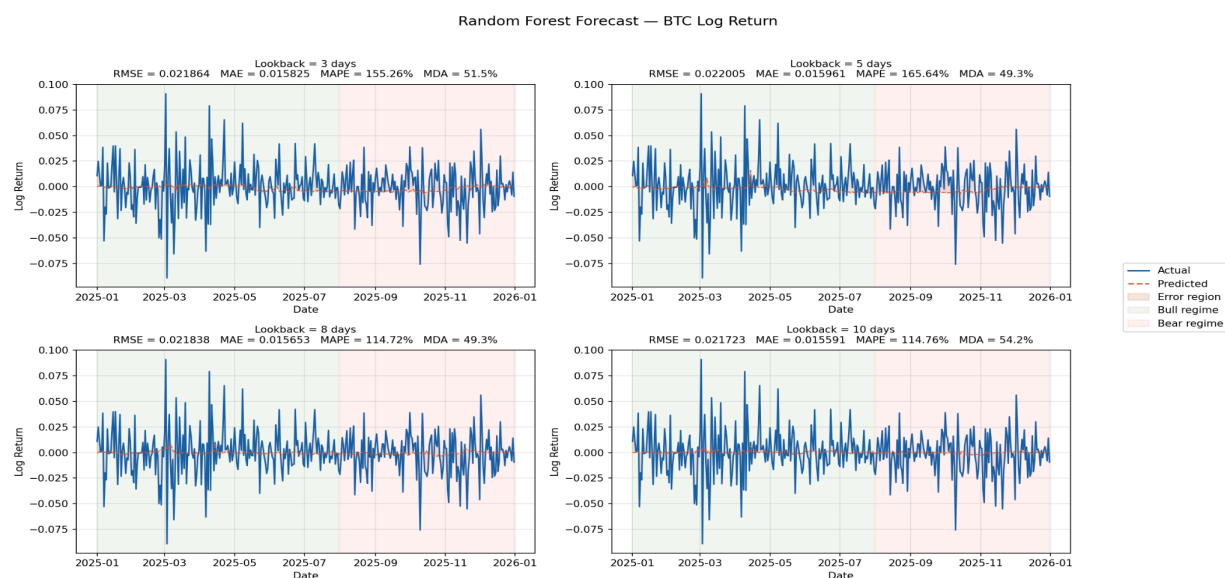
return series. None of the return results were very meaningful so attached we will show all lookbacks for all models.



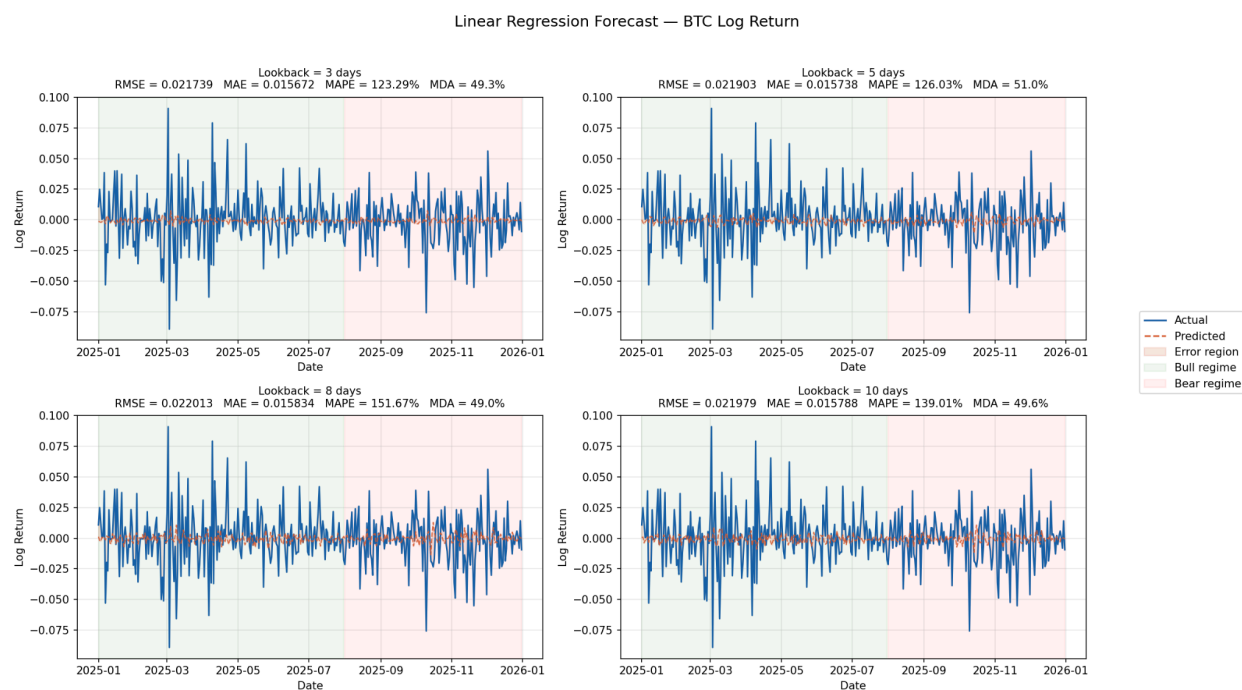
GRU had almost identical results, for near random targets the choice of gating mechanism makes no difference.



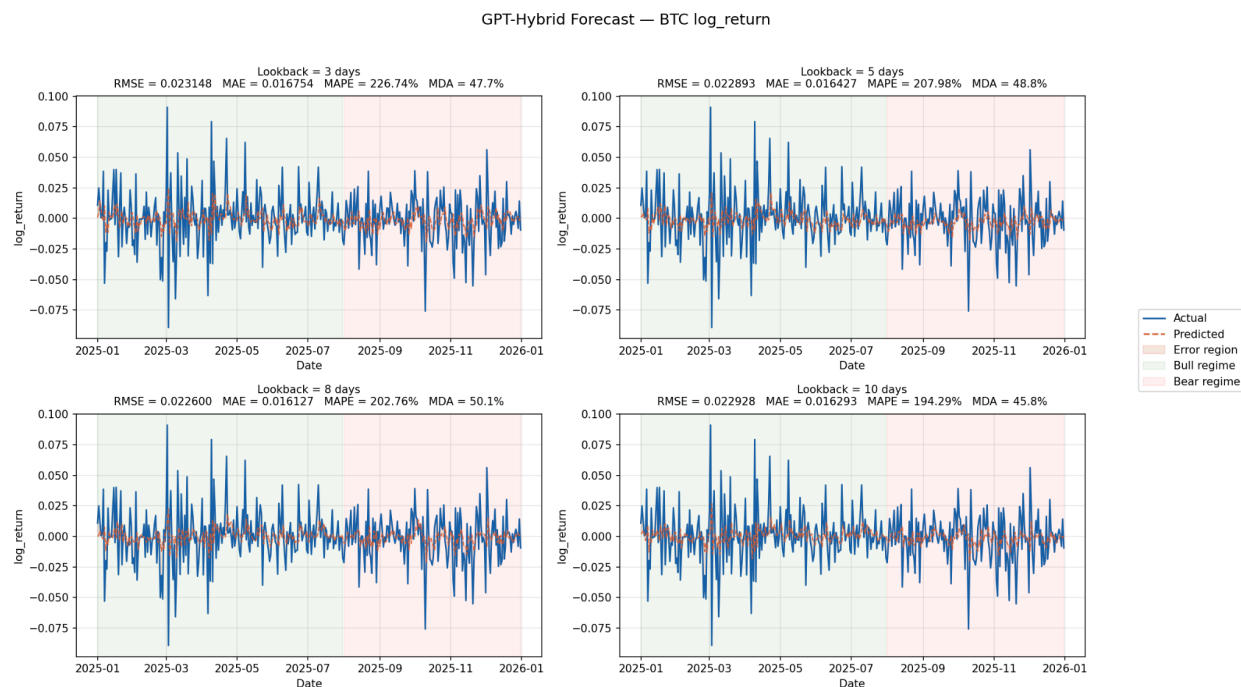
RF got to an RMSE of 0.02172 at lookback 10 and MDA 54.2%, a relatively significant directional edge over the deep learning models, potentially hinting at the fact that classic ML models could capture some directional signal the recurrent models miss.



LR RMSE was 0.02174 at lookback 3, MDA 49.3%, flat predictions near zero at all lookbacks, no directional edge, basically random guessing.



GPT Hybrid achieved its best RMSE at lookback 8 with 0.02260 and MDA of 50.1% which is slightly weaker magnitude accuracy than the RF while achieving a random directional score, suggesting the hybrid structure adds no advantage for white noise return series.

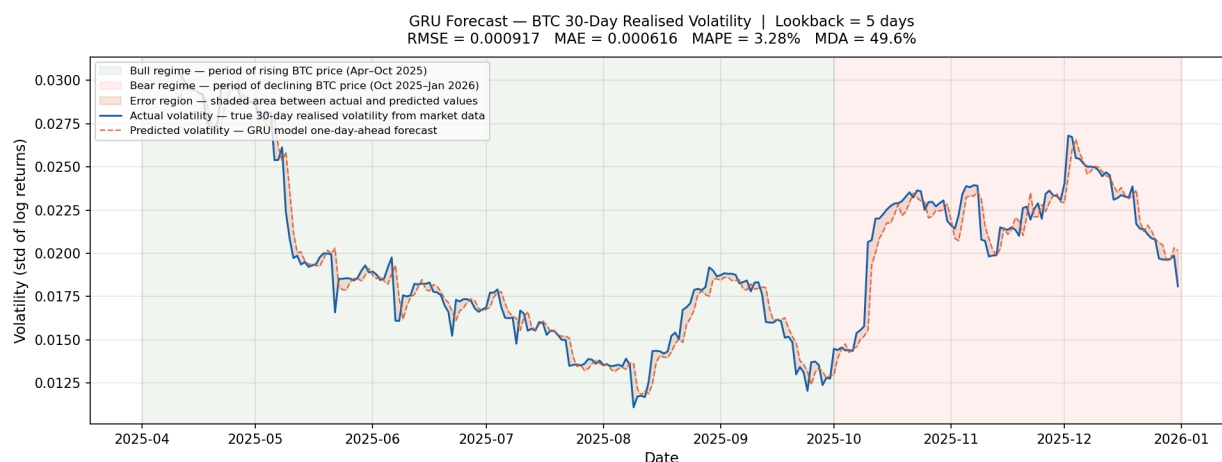


Volatility Forecasting

Our most predictable target due to clustering which you will read about plenty in part 5. LSTM achieved RMSE of 0.001441 at lookback 5 and MAPE 6.07%, MDA 47.7%, predicted line tracks volatility well as with all models here.



GRU had better results than LSTM, consistent with the expectation of GRU having simpler architecture with fewer parameters allowing it to generalise better on auto correlated series. RMSE at 0.000917, MAPE 6.07%, MDA 49.6%.

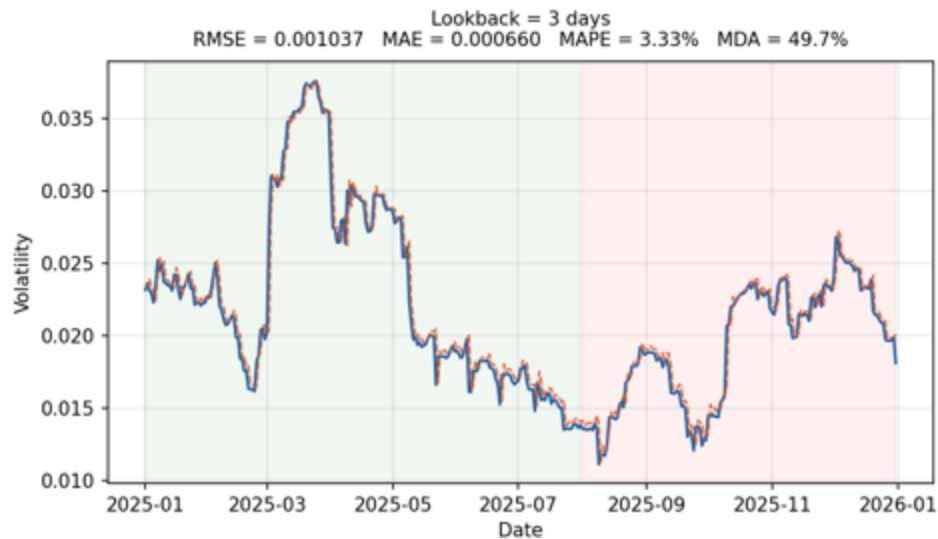


RF had RMSE at 0.001648, MAPE 6.55% and MDA 53.3% at lookback 5. Achieved the highest directional accuracy reflecting the strength of the ensemble at recognising the clustering pattern shown by volatility.

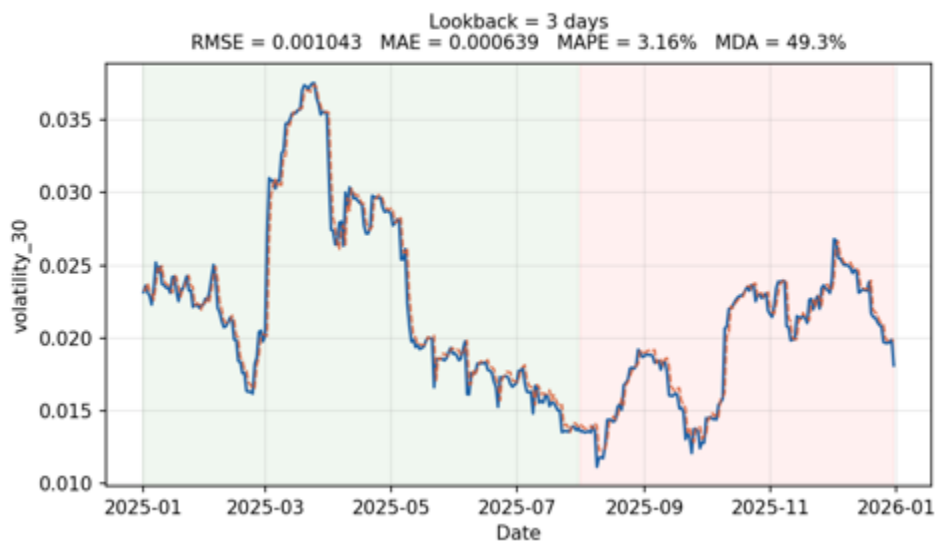


LR exemplified a strong result at lookback 3, RMSE of 0.001037, MAPE 3.33%, MDA 49.7%. Highlights the strong linear auto correlation pattern in 30 day volatility.

Linear Regre

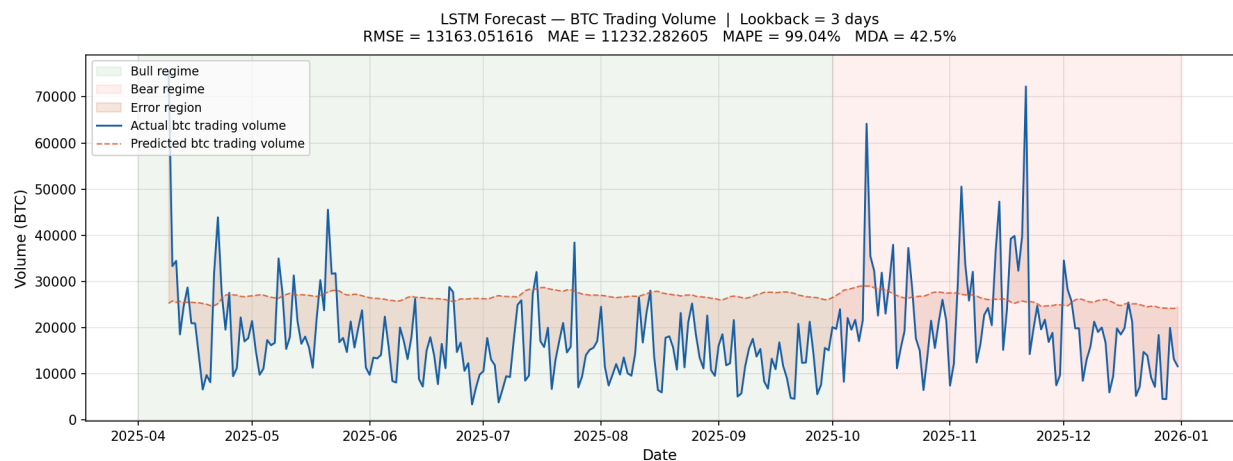


GPT effectively matching the LR, the linear aspect of the hybrid dominates and the added complexity has little benefit.

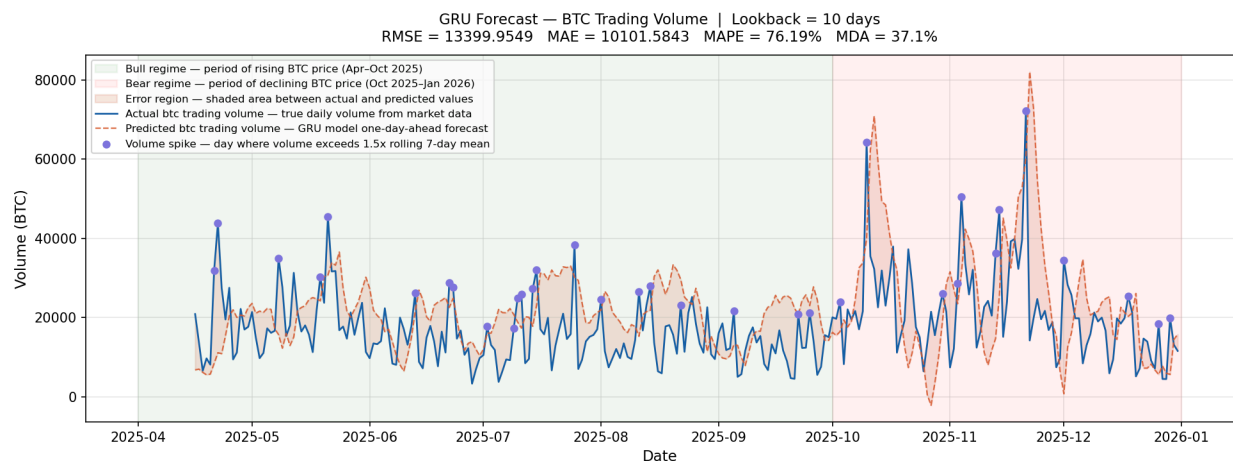


Volume Forecasting

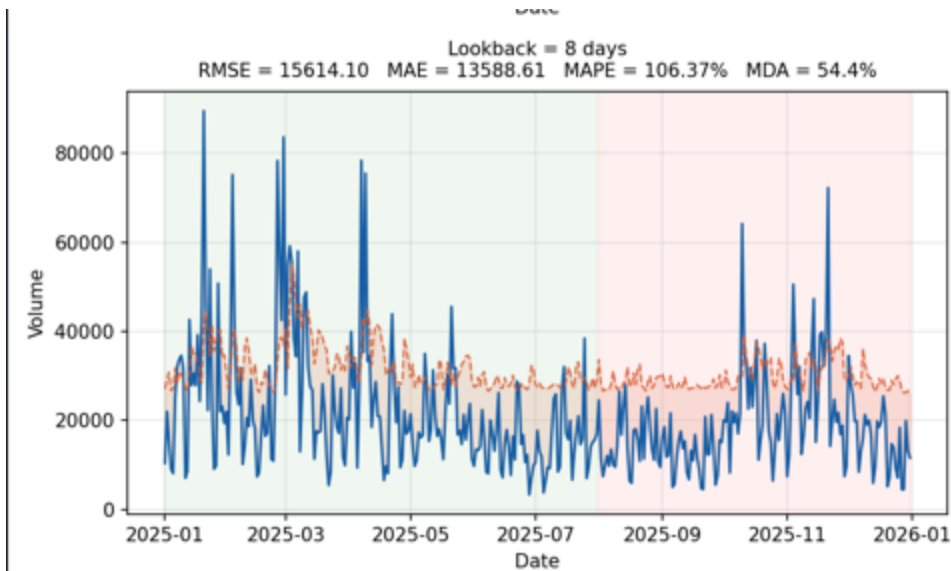
Extremely difficult to forecast because it is impacted by things outside of our dataset. LSTM's strongest result came at lookback 3 with an RMSE of 13163 btc and MDA at 42.5%. The model could roughly learn the average volume level but could not anticipate spikes and at longer lookbacks the results only worsened.



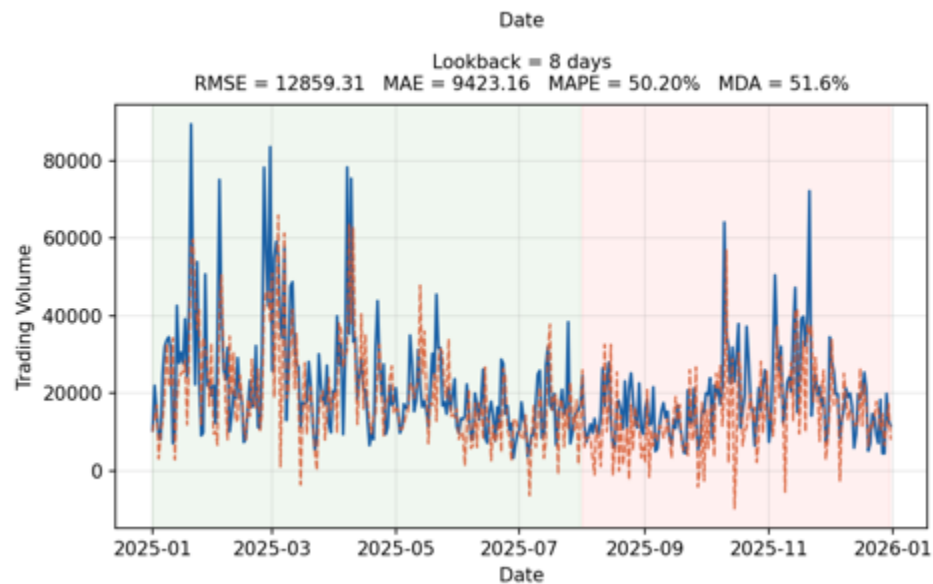
GRU had a comparable short lookback performance to LSTM, however at the longer lookback it could somewhat follow the spikes. At lookback 10 you can see the predicted line generally does a good job. The GRU's simpler gating mechanism is less susceptible to the overfitting and divergence that caused the LSTM to get its prediction so wrong.



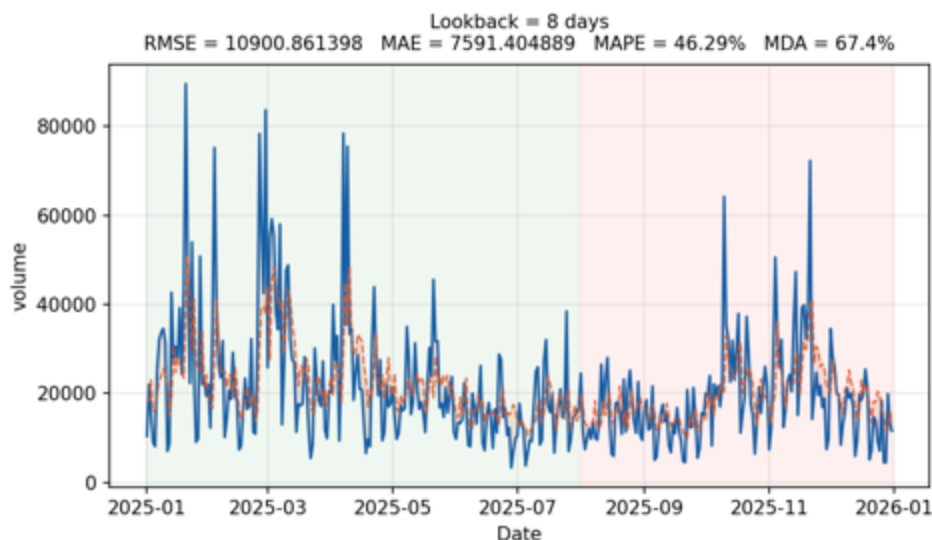
Random Forest achieved its best RMSE of 15,614 BTC at lookback 8 with MDA of 54.4%, weaker magnitude accuracy than the deep learning models at short lookbacks but achieving more stable and consistent directional performance across all lookback windows.



LR also had its best result at lookback 8 with RMSE at 12859 btc and MDA at 51.6%, this leads us to the conclusion that for spike dominated series a simpler linear trend model is more stable than complex architectures.



GPT, the only model to consistently track volume movements rather than predicting a flat average, with directional accuracy above 63% at every lookback window, indicating that the hybrid architecture's combination of structural and learned components provides a meaningful advantage for this challenging and irregular target. RMSE of 10,901 BTC at lookback 8 and MDA of 67.4%



5.0 Findings

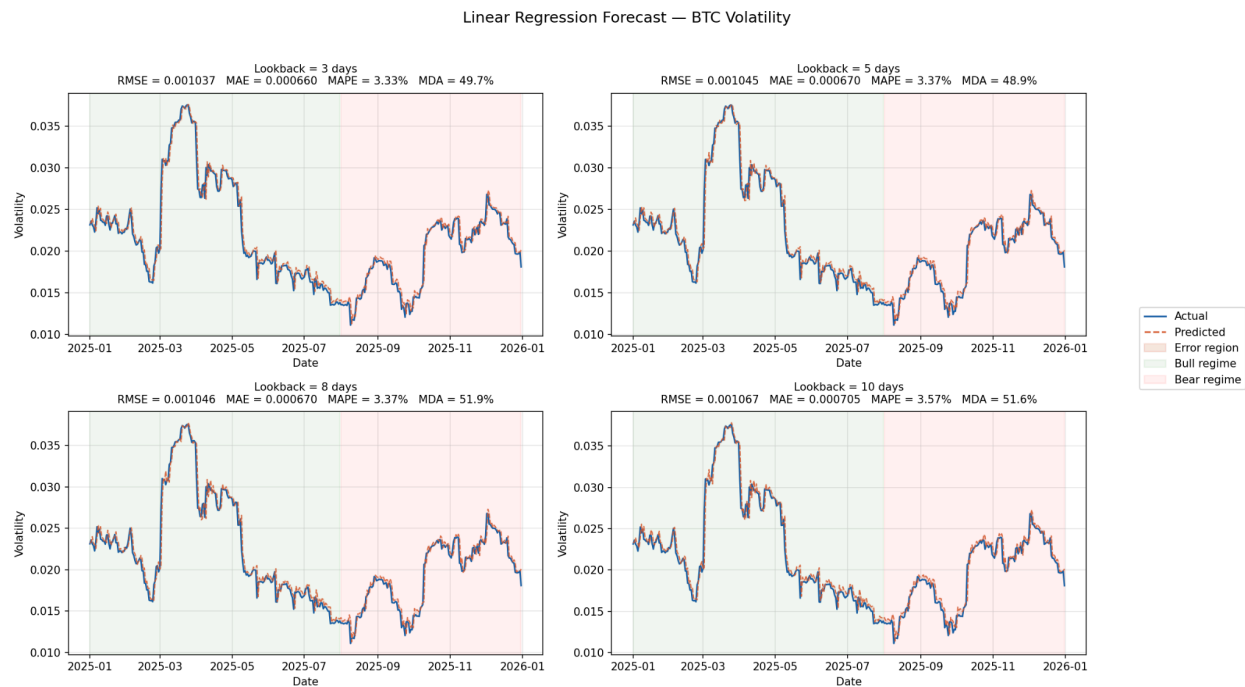
This section presents the findings and insights obtained from our prediction results, and what can be learnt and expected from predicting these features. The results are analysed in relation to how effectively different machine learning approaches, algorithms, and models could forecast cryptocurrency market behaviour using historical data alone. The findings compare and discuss the performance of the traditional, deep learning, and an extra GPT-hybrid, models across the four selected features in multiple lookback windows.

5.1 Volatility - The Most Predictable Target

Volatility was consistently the most predictable target across all architectures and lookback windows. Unlike price, returns, and trading volume, volatility displayed persistent clustering behaviour, allowing the models to identify and follow broader changes in market instability over time. Similar forecasting behaviour was observed across both Bitcoin and Ethereum, suggesting that volatility contained a more stable and learnable signal than the other targets explored throughout the project.

Linear Regression achieved the strongest overall volatility forecasting performance, recording the lowest RMSE values across all lookback windows. The best result was achieved at the 3-day lookback window with an RMSE of 0.001037 and a MAPE of 3.33%, while performance remained highly stable across the 5, 8, and 10 day configurations. The model was able to closely follow the overall volatility structure despite its simplicity, indicating that volatility patterns within cryptocurrency markets

contain strong autocorrelated behaviour that can be captured even by relatively simple traditional machine learning approaches.



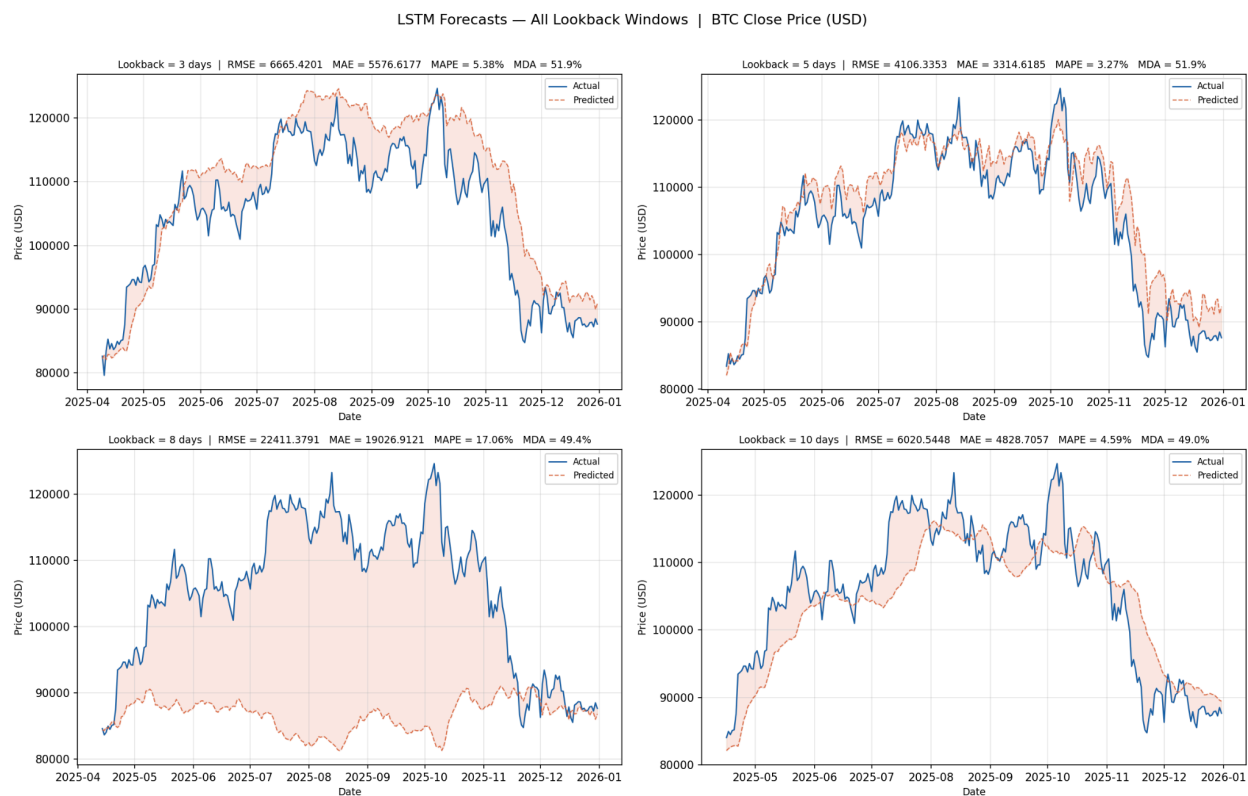
The findings suggest that cryptocurrency volatility contains persistent behavioural structure that remains partially predictable even within highly noisy and non-stationary market conditions. While the models still struggled with short-term directional changes and sudden turning points, they were able to identify broader patterns of market instability more effectively than with any other target. This indicates that volatility may provide more useful forecasting information than raw returns or trading activity when relying on historical cryptocurrency data alone.

5.2 Price - Trend Tracking, Not Direction

The close price results showed that cryptocurrency prices were partially predictable, but mainly at the broader trend level. Similar behaviour was observed across both Bitcoin and Ethereum, with the models generally able to follow overall upward and downward market movement over time. However, this did not translate into reliable short-term directional prediction, as the models frequently struggled around sudden reversals and rapid market shifts.

This behaviour can be clearly observed in the LSTM price forecasting results, where the model was able to follow the broader structure of the market while still showing noticeable limitations around turning points and sharp price movements. Similar behaviour was also present within the Linear Regression and Random Forest models,

suggesting that the models were learning general market structure rather than consistent daily directional behaviour.

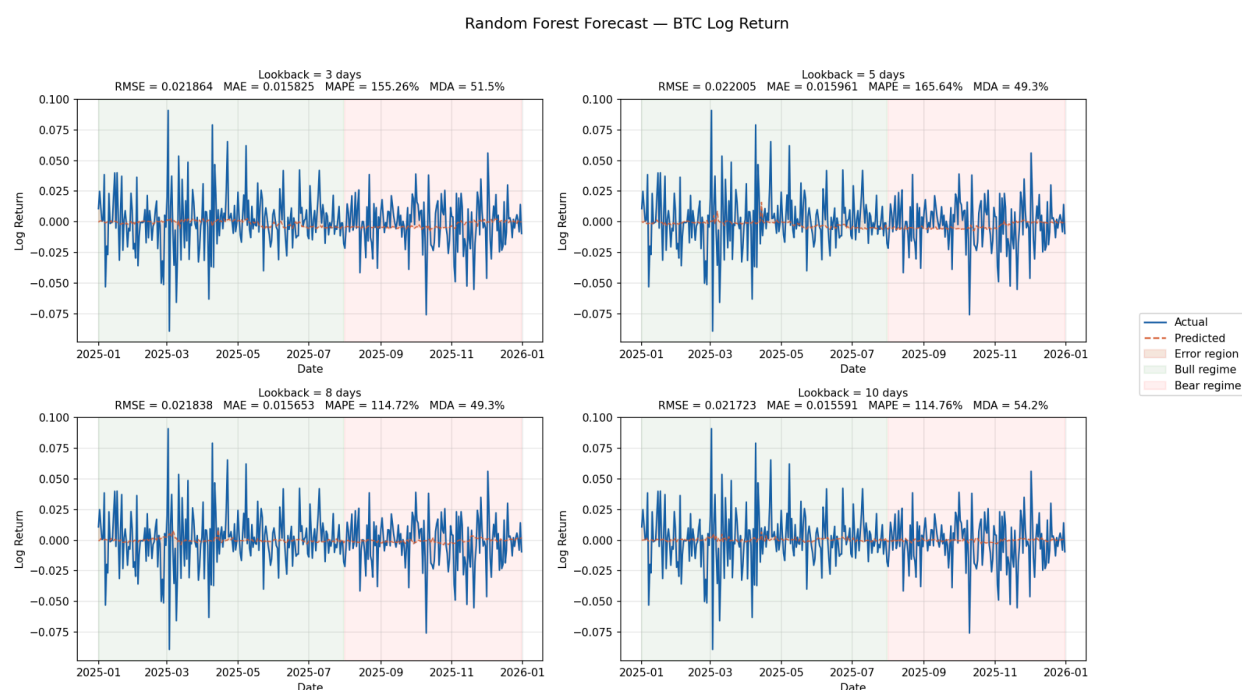


The findings suggest that cryptocurrency price series contain some learnable long-term structure, particularly during gradual market trends, but far less reliable information at the short-term directional level. While the models were capable of recognising broader market movement, they struggled whenever market behaviour became heavily influenced by sudden external events or rapid sentiment changes. This indicates that historical price data alone may be sufficient for identifying general market direction over time, but remains limited for producing reliable short-term forecasting or actionable day-to-day trading signals within highly volatile cryptocurrency markets.

5.3 Return - Near White Noise

Log return forecasting revealed that cryptocurrency returns were the least stable and most unpredictable target throughout the project. Similar behaviour was observed across both Bitcoin and Ethereum, with all architectures struggling to identify meaningful and consistent patterns within the return series. Unlike volatility and close price forecasting, the models were unable to reliably capture the rapid directional changes present within daily returns.

This behaviour can be clearly observed in the Random Forest return forecasting results, where the predictions remained relatively flat while the actual return series fluctuated rapidly between positive and negative values. Similar behaviour was also present across the Linear Regression and LSTM models, suggesting that the issue was not related to a specific architecture, but rather to the nature of the return series itself.



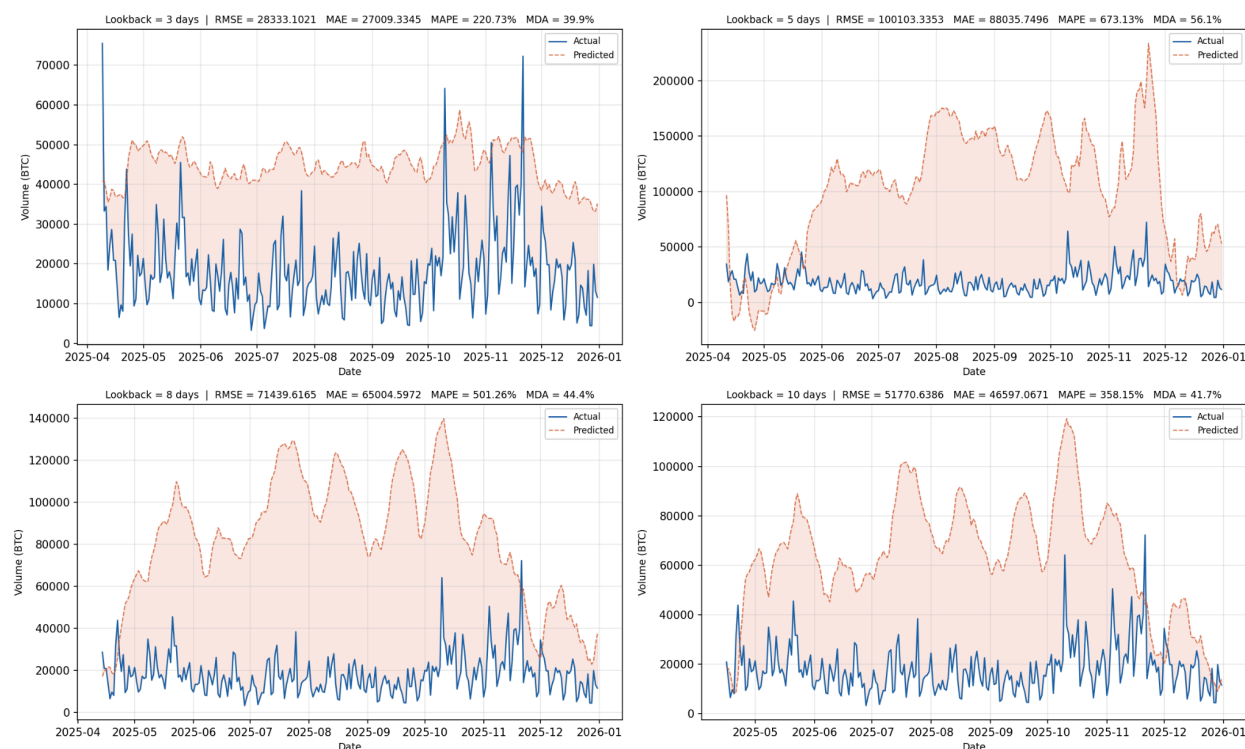
The findings suggest that short-term cryptocurrency returns behave similarly to near white noise, containing very limited learnable structure when relying only on historical market data. Sudden market reactions, investor sentiment shifts, news events, and other external factors appeared to dominate return behaviour, making consistent prediction highly difficult across all models and lookback windows.

5.4 Volume - Spike-Driven, Hardest Target

Trading volume forecasting produced the most unstable behaviour across all architectures and lookback windows. Similar patterns were observed across both Bitcoin and Ethereum, with all models struggling to accurately capture the sudden spikes and irregular movements present within the volume series. Unlike volatility and close price forecasting, trading volume displayed highly inconsistent behaviour that was heavily influenced by abrupt market activity.

This behaviour can be clearly observed in the LSTM trading volume forecasting results, where the model frequently overestimated market activity and produced large deviations from the actual volume series. Similar issues were also present within the Linear Regression and Random Forest models, which often generated overly smooth predictions that failed to react appropriately to sudden spikes in trading activity.

LSTM Forecasts — All Lookback Windows | BTC Trading Volume



The findings suggest that cryptocurrency trading volume is highly event-driven and contains limited stable structure when relying only on historical market data. External factors such as news events, liquidations, investor sentiment, and sudden increases in market participation appeared to influence trading activity in ways that were not captured by the feature set used throughout the experiments. As a result, trading volume emerged as the hardest target to forecast consistently across all modelling approaches.

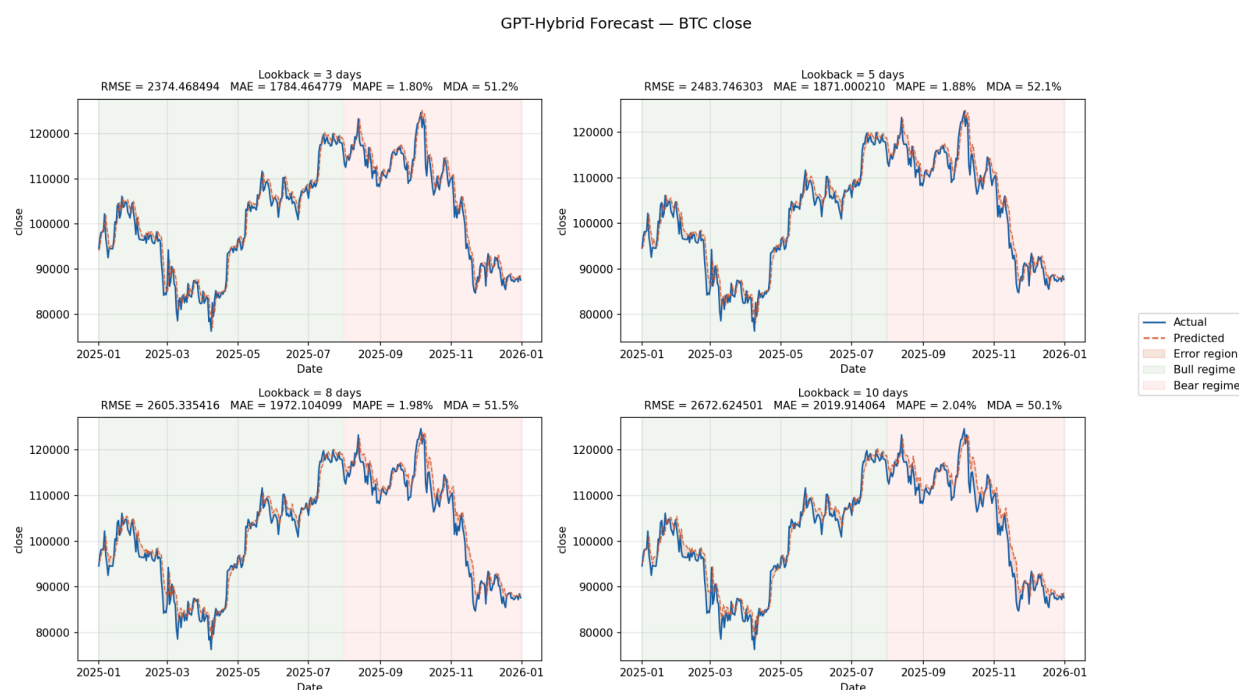
5.5 Cross-Model Comparison & Bonus ChatGPT-Hybrid Benchmark

As a bonus addition to the project, an AI-generated hybrid prediction model was introduced alongside the discussed traditional ML and deep learning models. The decision to include this model was based on the growing popularity and increased accessibility and use of AI-assisted chatbots and AI-Gens, particularly within finance

and data analytics, taking this in consideration, the ChatGPT-hybrid model was added to explore how an AI-generated forecasting approach would perform within cryptocurrency prediction tasks. By solely user-prompting on ChatGPT, the AI generated a hybrid model which combines Naive Last, Lag Mean, Ridge Regression, and Histogram Gradient Boosting predictions into a single forecasting framework.

5.5.1 Price

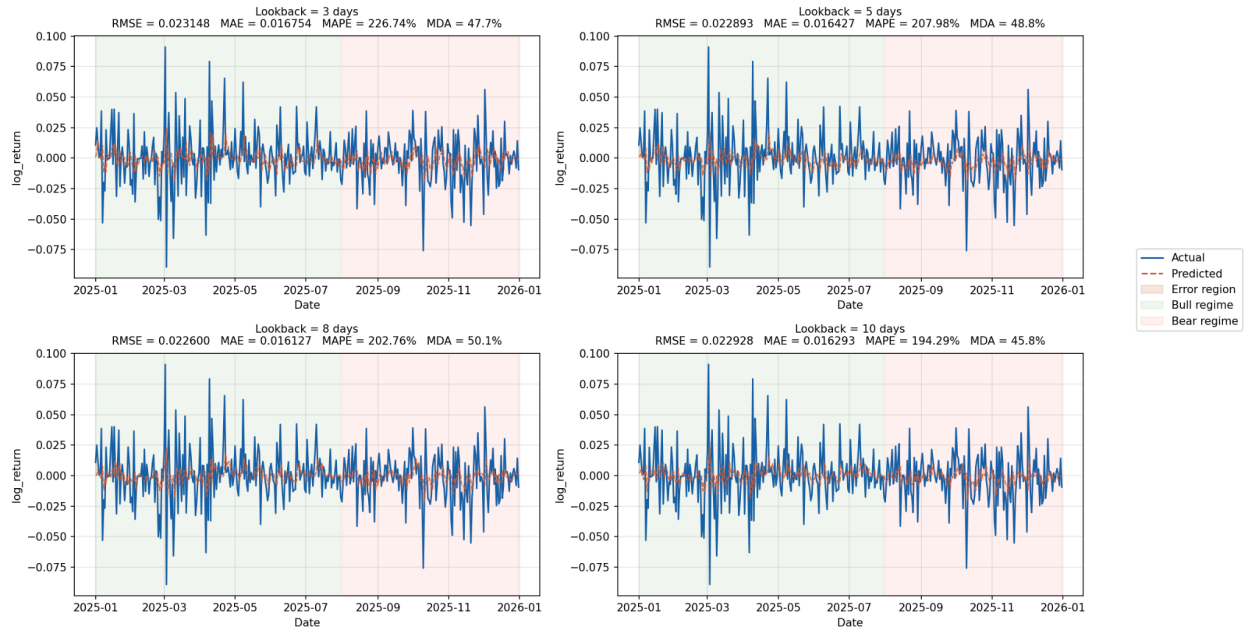
The ChatGPT-hybrid model performed strongly when forecasting close price trends, particularly during gradual upward and downward market movements. Similar to the traditional ML and deep learning models, it was generally able to follow the broader structure of the market while still struggling during sudden price shocks and rapid reversals. Overall, its close price forecasting performance was competitive with several of the other architectures explored throughout the project.



5.5.2 Return

Similar to the other architectures, the ChatGPT-hybrid model struggled substantially when forecasting log returns. The predictions frequently failed to align with the rapid directional changes present within the actual return series, reinforcing the finding that cryptocurrency returns behaved similarly to near white noise and contained limited predictable structure.

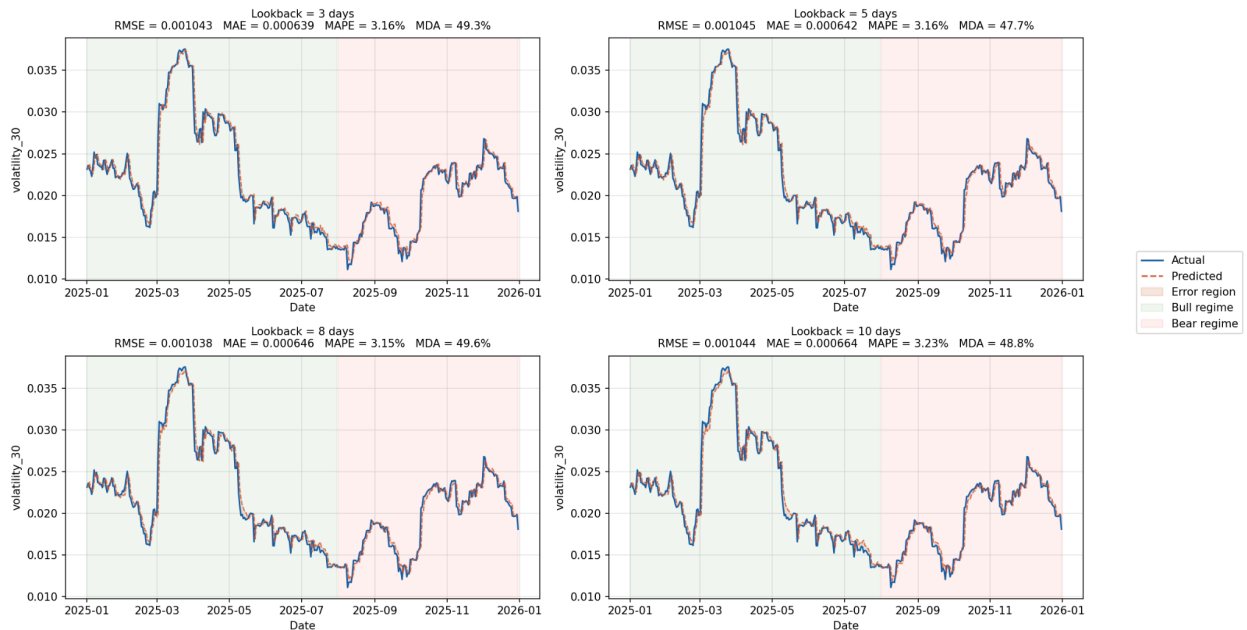
GPT-Hybrid Forecast — BTC log_return



5.5.3 Volatility

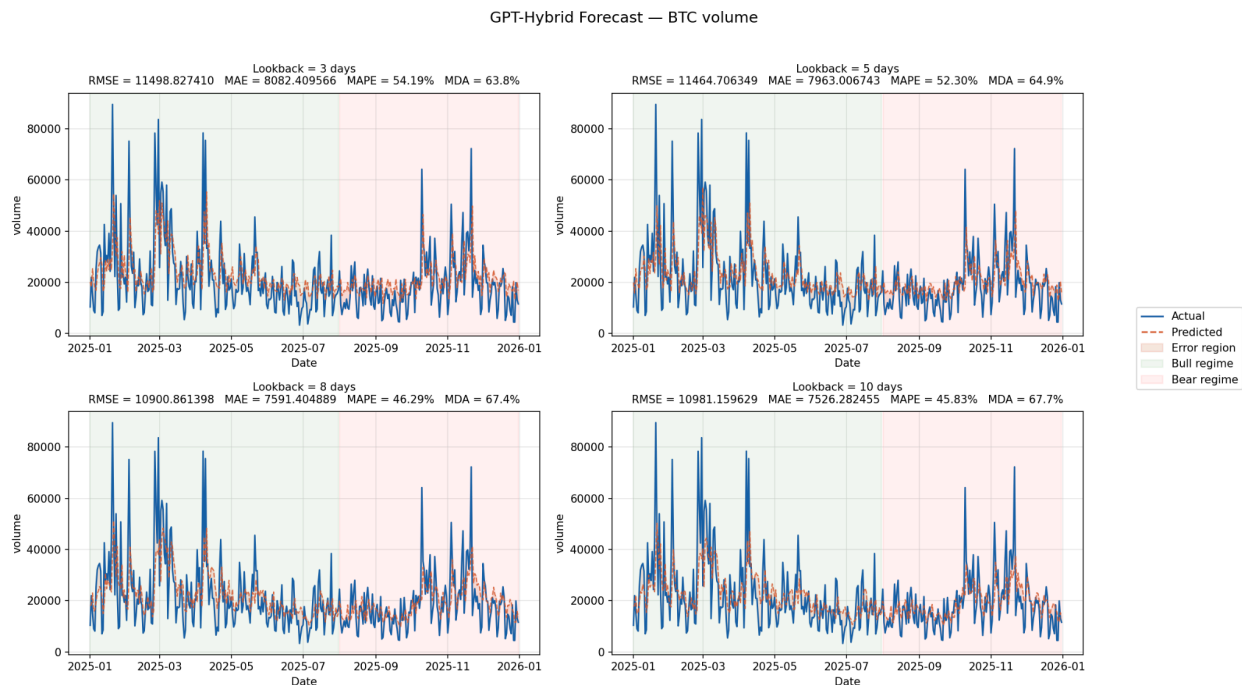
The hybrid model performed particularly well when forecasting volatility, closely following periods of changing market instability and major volatility regimes. Its behaviour was highly consistent with the findings observed throughout the project, further reinforcing the idea that volatility contained the strongest learnable structure among all targets.

GPT-Hybrid Forecast — BTC volatility_30



5.5.4 Volume

The ChatGPT-hybrid model produced comparatively stronger trading volume forecasts than expected, particularly during periods of increased market activity and sudden spikes. While the model still struggled with highly irregular movements, it handled spike-driven behaviour more effectively than some of the other architectures in several configurations, making it one of the more competitive approaches for volume forecasting within the project.



The ChatGPT-hybrid model has produced results that were generally consistent with the findings and insights observed from the traditional and deep learning models. While it did not overcome the unpredictability associated with cryptocurrency forecasting, AI-generation has demonstrated to be capable of producing forecasting frameworks competent enough to produce reasonably reliable results for targets that were naturally more predictable, particularly volatility and general price trends.

6.0 Recommendations

6.1 Add External Data Sources (Sentiment, On-Chain, Macro)

The most important recommendation from this project is to expand the modelling framework beyond historical price and volume data. Our models were built using OHLCV market data and engineered technical indicators. This was appropriate for testing the limits of price-history-only forecasting, but the results showed that this data source has clear limitations. Volatility was the most predictable target because volatility clustering gave the models a persistent pattern to learn. However, log return and trading volume were much harder to forecast because their largest movements were often caused by events that were not captured in the historical price series.

Future research should therefore include external data sources such as news sentiment, social media sentiment, on-chain activity, exchange inflows and outflows, liquidation data, funding rates, open interest and macroeconomic indicators. These features are especially relevant for return and volume prediction. For example, a sudden increase in trading volume may not be predictable from the previous 3, 5, 8 or 10 days of volume alone, but it may be easier to explain if there is also unusual exchange flow activity, liquidation pressure, or strong news sentiment at the same time.

This recommendation does not mean that external data will make cryptocurrency fully predictable. Rather, our results suggest that historical price data alone has a clear boundary. It contains useful signals for volatility, limited signals for price level, and weak signals for return direction and volume spikes. The next stage of work should test whether this boundary improves when market context is added. This would help the Data Science Institute determine whether poor return and volume performance was caused by model limitations, missing information, or both.

6.2 Match Model Architecture to Target Variable

A consistent finding across the project was that no single model configuration worked best across all four target variables. Each target behaved differently, which meant that each required a different modelling approach. Close price, log return, 30-day rolling volatility and trading volume should therefore be treated as separate forecasting problems rather than as one general cryptocurrency prediction task.

Volatility was the strongest target because it had a clearer time-series structure. High-volatility periods tended to continue for some time before reducing again, which gave the models a more stable pattern to learn. For this reason, volatility forecasting

worked better with more conservative configurations that preserved this smoother autocorrelated signal. Price forecasting was different. The models could track the broad price level because price is highly autocorrelated, but this did not mean they could predict direction. This was shown by Mean Directional Accuracy staying close to random, even when the price line appeared visually close to the actual values.

Log return was the least suitable target for price-history-only modelling because most predictions collapsed towards zero. This reflected the difficulty of finding a reliable daily return signal in a noisy market. Volume was also highly unstable because it was driven by sudden spikes. At longer lookback windows, some LSTM volume models produced negative predictions or overestimated spikes by 5 to 10 times.

The recommendation is therefore to match architecture and hyperparameters to the target variable. Volatility models should prioritise stability, price models should be evaluated carefully for directional skill, return models should not be over-engineered without external features, and volume models should include constraints that prevent unrealistic outputs.

6.3 Use the Right Metric per Target (RMSE + MDA)

The evaluation framework should continue to use multiple metrics because each metric explains a different part of model performance. RMSE is useful because it measures the average size of prediction errors and penalises larger mistakes more strongly. This makes it suitable for targets such as close price, volatility and volume, where large errors are important. MAE is also useful because it gives a more direct average error and is less affected by extreme values than RMSE.

MAPE should be used more carefully. Although it is easy to interpret because it expresses error as a percentage, it becomes less reliable when actual values are close to zero. This is especially relevant for log return, because returns are often centred around zero. It can also create misleading results for volume when actual values are low or when sudden spikes occur. For these reasons, MAPE should not be treated as the main metric for every target.

Mean Directional Accuracy is important because it measures something that RMSE, MAE and MAPE do not. It shows whether the model correctly predicts whether the target will move up or down. This mattered in our results because some price models appeared useful when judged by error size, but their directional accuracy remained close to 50 percent. This means the models were mostly tracking the level of the series rather than predicting useful directions.

For future DSI work, RMSE and MDA should be reported together as the minimum evaluation requirement. RMSE should be used to assess magnitude accuracy, while

MDA should be used to assess directional skill. This prevents future models from being overstated when they only follow price levels without producing meaningful forecasting value.

6.4 Integrate into the DSI Research Workflow

The project should be integrated into the Data Science Institute's workflow as a reusable research pipeline rather than a deployed trading system. The main value of the project is that it provides a structured way to test where a predictive signal exists across different cryptocurrency targets. This can help DSI decide which forecasting tasks are worth further investment and which tasks require more data before model performance can realistically improve.

The pipeline should be maintained in modular form, with separate components for data collection, feature engineering, model training and evaluation. This allows future researchers to update one part of the workflow without rebuilding the full project. For example, new assets can be added to the data collection stage, new sentiment or on-chain indicators can be added to the feature stage, and new architectures can be tested against the existing models. The current Linear Regression, Random Forest, LSTM, GRU and GPT-hybrid results should be kept as benchmarks for future comparison.

A common experiment tracking process should also be used. Each experiment should record the dataset version, asset, target variable, lookback window, model architecture, hyperparameters, scaler, training period, test period and evaluation metrics. This directly addresses one of the main project issues, where early experiments were sometimes duplicated or difficult to compare because results were not tracked consistently.

Finally, the workflow should support walk-forward retraining. Since cryptocurrency markets are non-stationary, a model trained once on historical data should not be assumed to remain reliable. Regular retraining and out-of-sample monitoring would make the pipeline more useful for ongoing DSI research.

7.0 Deployment, Future Work and Reflection

7.1 Deployment Considerations & Constraints

Any deployment of this project must begin with a clear limitation: this is not a trading system. The project does not include portfolio construction, transaction costs, slippage, execution rules, risk limits or live trading decisions. Instead, it should be understood as

a research system that tests which cryptocurrency variables contain useful predictive signals when only historical market data is used.

The biggest deployment constraint is non-stationarity. The models were trained on data from 2021 to 2024 and tested on 2025 data. However, the 2025 market period contained price levels and behaviour that were not fully represented in the training period. This created problems for model generalisation. One clear example was the MinMaxScaler issue. The scaler was fitted on the training data, but when 2025 prices exceeded the training range, scaled values moved outside the expected range. This caused some model predictions to be compressed towards the upper end of the training distribution and contributed to under-prediction during large price movements.

There are also practical modelling constraints. LSTM and GRU models were highly sensitive to target-specific hyperparameters. Batch normalisation also created issues when the final training batch contained only one sample, requiring the data loader to drop incomplete batches. Volume models were especially unstable at longer lookback windows, sometimes producing negative values or very large overestimates.

For these reasons, any deployment should include strict monitoring and validation. Forecasts should be checked for unrealistic outputs, model drift should be tracked over time, and retraining should occur regularly. The model outputs should be treated as research evidence, not as automatic decision signals.

7.2 Integration into Business Information Systems

The most practical way for DSI to use this project is as a research dashboard and model comparison system. The pipeline can help researchers compare assets, targets, lookback windows and model architectures in a consistent way. This would make the project useful as part of a wider business information system, even though it should not be connected directly to trading execution.

The system could be integrated through scheduled data collection, automatic feature engineering, model training, evaluation and reporting. New cryptocurrency data could be collected daily or weekly, processed through the existing feature pipeline, and tested against stored model configurations. The results could then be shown in a dashboard that reports RMSE, MAE, MAPE and MDA for each asset, target variable, model and lookback window. This would allow researchers to quickly see which targets are currently performing well and which are degrading under new market conditions.

The dashboard should also include warning indicators. For example, if MDA stays close to 50 percent, the system should make clear that the model does not have useful directional skill, even if RMSE looks strong. If new price values exceed the range used by the scaler, the system should flag possible scaling distortion. If a volume model

produces negative or unusually large predictions, the system should identify those outputs as invalid or unstable.

This kind of integration would support better research decisions. DSI could use the system to monitor model performance, compare new architectures against existing baselines, and decide whether further forecasting work is justified for a particular target. The purpose would be to improve research visibility and reproducibility, not to automate financial decisions.

7.3 Future Work & Research Extensions

Future work should focus on testing whether predictability improves when the models are given more information and evaluated under more realistic conditions. The first research extension should be the addition of external data. News sentiment, social media activity, on-chain metrics, exchange inflows and outflows, liquidation data, funding rates and macroeconomic indicators should be tested alongside the existing OHLCV features. These variables are especially relevant for log return and trading volume, where the current models performed poorly because many movements were driven by external events.

The second extension should be more advanced model comparison.

Transformer-based time-series models and Temporal Fusion Transformers could be tested against the current Linear Regression, Random Forest, LSTM, GRU and GPT-hybrid models. However, these models should only be considered improvements if they produce stronger out-of-sample results, especially in Mean Directional Accuracy. Adding complexity alone is not enough. The project showed that deep learning models are not automatically better and can become unstable if their configuration does not match the target variable.

The third extension should be walk-forward retraining. Instead of relying on one fixed split between 2021 to 2024 training data and 2025 test data, future work should simulate a live forecasting environment where models are repeatedly retrained as new data becomes available. This would better reflect the way cryptocurrency markets change over time.

Future work could also test additional assets beyond Bitcoin and Ethereum, compare daily and intraday forecasting horizons, and use target-specific loss functions. For volume, future models should include non-negative output constraints or spike-aware methods. For returns, classification models may be more suitable than regression if the main interest is direction.

7.4 Lessons Learned

The main lesson from this project is that cryptocurrency forecasting cannot be judged as one simple prediction problem. Each target variable had a different level of predictability. Volatility was the most predictable because it had persistence over time. Price could be tracked at the level because today's price is usually close to recent prices, but this did not mean the models could predict direction. Log returns were close to white noise, and volume was difficult because large spikes were usually caused by external events.

The project also showed that complex models are not automatically better. LSTM and GRU models required careful target-specific tuning, and their performance changed significantly depending on the chosen hyperparameters. In some cases, aggressive configurations improved responsiveness for price but damaged the smoother structure needed for volatility. In other cases, volume models became unstable at longer lookbacks. This showed that model design needs to be guided by the behaviour of the target variable rather than by the assumption that a more advanced architecture will perform better.

Another lesson was that preprocessing decisions strongly affect the validity of the results. The first 30 rows had to be removed because rolling-window features created NaN values. This had to be done before the chronological train-test split to avoid misalignment. The MinMaxScaler issue also showed that common preprocessing tools can fail when financial data moves into a new regime.

The group also learned that experiment tracking is critical in machine learning projects. Early duplicate experiments and inconsistent result recording created unnecessary work. Once a shared tracking process was introduced, results became easier to compare and validate.

7.5 Group Reflection & Closing

Overall, the project gave the group a clearer understanding of the limits of machine learning in financial forecasting. The main value of the work was not producing a model that could reliably trade cryptocurrency, but showing which parts of the market contain stronger predictive structure and which parts remain highly uncertain. Volatility produced the clearest signal, while log returns and trading volume showed the limits of using historical market data alone.

As a group, we also learned that modelling decisions need to be tied closely to the target variable. A model configuration that works for volatility may not work for returns or volume, and a visually close price forecast can still have limited directional value. This reinforced the importance of using multiple evaluation metrics and avoiding overstated claims.

The final outcome is a reusable forecasting pipeline that DSI can build on for future research. It provides a clear benchmark across assets, models, targets and lookback windows, while also identifying where extra data, stronger tracking and walk-forward retraining would be needed. This makes the project useful as a research foundation, rather than a finished deployment system.

8.0 Difficulties Encountered

8.1 Modelling & Architecture Difficulties

The biggest modelling problem came from the extreme sensitivity of LSTM and GRU architecture to hyperparameters depending on what variable was being modeled. For price and volume we needed aggressive hyper parameters (such as; 128 hidden units, one layer, stacked network, OneCycleLR scheduler, and batch norm regulariser) to achieve acceptable results. for volatility, however using the same hyper parameters destroyed the long term auto-correlated signal that is key to volatility becoming learn-able. This meant that a much more conservative set of hyperparameters was needed (smaller hidden units, no learning rate scheduled, and a slower ReduceLROnPlateau schedule).

Determining the "correct" set of hyperparameters for each target variable required considerable amounts of experimentation. The amount of experimentation far exceeded original expectations for the modeling phase. Volume proved to be the most unstable. At longer look back windows, LSTMs would either collapse to predictably wrong (physically impossible) negatives for volume or they would produce an estimate that was anywhere from 5 to 10 times too large because it was simply over-fitting to spikes in the high volume of the training data. GRUs were brought into use partially to help solve some of the issues with volume. GRUs have fewer parameters than LSTMs and are less likely to over-fit. GRUs did improve performance against volume, but volume was still the hardest target to model across all possible model configurations and all possible look-back window sizes.

A second major issue that surfaced was how batch normalization interacts with small training batches. When batch normalization encounters a single sample batch it cannot compute useful statistics. This caused silent training failures until the training data loader was modified to include last-batch drop-out.

8.2 Data, Feature & Non-Stationarity Issues

The feature engineering pipeline created NaN values within the first 30 rows of every dataset due to the maximum rolling window size used to calculate volatility and indicators. Therefore, removing those rows before creating sliding windows required careful testing to ensure that the removal occurred before splitting into chronological train/test sets because if done otherwise (i.e., after the split), it could have caused some level of index misalignment which would have been detrimental to the validity of the results obtained through evaluating.

In addition to this, there were some structural issues with how the MinMaxScaler was utilized on the data. The MinMaxScaler was fitted based upon training data in the years 2021-2024, then applied to the test data in 2025. However, the prices in 2025 were significantly higher than the highest prices seen during the training periods. Specifically, the rapid increases in price in late 2025 resulted in many prices being outside of the scaling range of the MinMaxScaler. Consequently, when scaled and inverted back into original space, the price predictions made by models using such a scaler were compressed toward the top end of the training distribution, resulting in systematic under-prediction of price movement during the extreme movements of price seen in the test period. Any scaler fit historically will inherently fail when applied forward in time under conditions of non-stationarity.

Finally, empirical evidence throughout the project confirmed that cryptocurrency markets are non-stationary. Training loss reached levels that seemed reasonable given the training process; however, out-of-sample performance on the 2025 test set generally performed poorly relative to what had been demonstrated while training. These differences across the two market regimes were evident specifically for return and volume targets, wherein no model was able to achieve statistically significant directional accuracy over the random baseline.

8.3 Workload, Timeline & Coordination

The modelling load was greater than originally anticipated. Therefore, each model-target combination required individual hyperparameter optimisation (four look back periods per configuration), as a consequence of which there were many models produced from training data using LSTM/GRU architectures for both Bitcoin and Ethereum. Using only CPUs to run these experiments was very time consuming;

therefore, the availability of GPU-enabled computing varied throughout the project team and created an asymmetrical load of computational work.

In early stages of the modelling process, various team members worked on their own experiments in parallel but did not have a common repository to track results, thus causing duplicate runs and varying reports on experimental results. Furthermore, at times, experiments were unintentionally performed using previous versions of the dataset instead of the final cleaned version, thus producing results that could not be repeated after the errors were discovered. A common tracking log was developed during the mid-project to document the specific configuration of all models used in the model development process, including target variable and look back period. All data reported in Section 5 was validated based upon this log.

Additionally, the time taken for data preparation and validation of pipelines exceeded expectations thereby reducing the amount of time available for conducting systematic experimentation related to deep learning hyperparameter tuning. Additionally, it is possible that the reduced amount of time allowed for such experimentation contributed to the unsteady performance seen in several volume-based model configurations.

9.0 References

Akyildirim, E, Goncu, A & Sensoy, A 2021, 'Prediction of cryptocurrency returns using machine learning', *Annals of Operations Research*, vol. 297, no. 1, pp. 3-36, <https://doi.org/10.1007/s10479-020-03575-y>.

Bouri, E, Gupta, R, Tiwari, AK & Roubaud, D 2017, 'Does bitcoin hedge global uncertainty? Evidence from wavelet-based quantile-in-quantile regressions', *Finance Research Letters*, vol. 23, pp. 87-95, <https://doi.org/10.1016/j.frl.2017.02.009>.

Bouteska, A, Abedin, MZ, Hajek, P & Yuan, K 2024, 'Cryptocurrency price forecasting – A comparative analysis of ensemble learning and deep learning methods', *International Review of Financial Analysis*, vol. 92, article 103055, <https://doi.org/10.1016/j.irfa.2023.103055>.

Catania, L, Grassi, S & Ravazzolo, F 2019, 'Forecasting cryptocurrencies under model and parameter instability', *International Journal of Forecasting*, vol. 35, no. 2, pp. 485-501, <https://doi.org/10.1016/j.ijforecast.2018.09.005>.

Chen, W, Xu, H, Jia, L & Gao, Y 2021, 'Machine learning model for bitcoin exchange rate prediction using economic and technology determinants', *International Journal of Forecasting*, vol. 37, no. 1, pp. 28-43, <https://doi.org/10.1016/j.ijforecast.2020.02.008>.

Conrad, C, Custovic, A & Ghysels, E 2018, 'Long and short-term cryptocurrency volatility components: A GARCH-MIDAS analysis', *Journal of Risk and Financial Management*, vol. 11, no. 2, article 23, <https://doi.org/10.3390/jrfm11020023>.

Corbet, S, Hou, YG, Hu, Y, Larkin, C & Oxley, L 2020, 'Any port in a storm: Cryptocurrency safe-havens during the COVID-19 pandemic', *Economics Letters*, vol. 194, article 109377, <https://doi.org/10.1016/j.econlet.2020.109377>.

Corbet, S, Hou, YG, Hu, Y, Larkin, C, Lucey, B & Oxley, L 2022, 'Cryptocurrency liquidity and volatility interrelationships during the COVID-19 pandemic', *Finance Research Letters*, vol. 45, article 102137, <https://doi.org/10.1016/j.frl.2021.102137>.

El Montasser, G, Charfeddine, L & Benhamed, A 2022, 'COVID-19, cryptocurrencies bubbles and digital market efficiency: Sensitivity and similarity analysis', *Finance Research Letters*, vol. 46, article 102362, <https://doi.org/10.1016/j.frl.2021.102362>.

Fang, T, Su, Z & Yin, L 2020, 'Economic fundamentals or investor perceptions? The role of uncertainty in predicting long-term cryptocurrency volatility', *International Review of Financial Analysis*, vol. 71, article 101566, <https://doi.org/10.1016/j.irfa.2020.101566>.

Hajek, P, Hikkerova, L & Sahut, JM 2023, 'How well do investor sentiment and ensemble learning predict bitcoin prices?', *Research in International Business and Finance*, vol. 64, article 101836, <https://doi.org/10.1016/j.ribaf.2022.101836>.

Hochreiter, S & Schmidhuber, J 1997, 'Long short-term memory', *Neural Computation*, vol. 9, no. 8, pp. 1735-1780, <https://doi.org/10.1162/neco.1997.9.8.1735>.

Hyndman, RJ & Koehler, AB 2006, 'Another look at measures of forecast accuracy', *International Journal of Forecasting*, vol. 22, no. 4, pp. 679-688, <https://doi.org/10.1016/j.ijforecast.2006.03.001>.

Ibrahim, A, Kashef, R & Corrigan, L 2021, 'Predicting market movement direction for bitcoin: A comparison of time series modeling methods', *Computers and Electrical*

Engineering, vol. 89, article 106905,
<https://doi.org/10.1016/j.compeleceng.2020.106905>.

Khedr, AM, Arif, I, El Bannany, M, Alhashmi, SM & Sreedharan, M 2021, 'Cryptocurrency price prediction using traditional statistical and machine learning techniques: A survey', *Intelligent Systems in Accounting, Finance and Management*, vol. 28, no. 1, pp. 3-34, <https://doi.org/10.1002/isaf.1488>.

Kristjanpoller, W & Minutolo, MC 2018, 'A hybrid volatility forecasting framework integrating GARCH, artificial neural network, technical analysis and principal components analysis', *Expert Systems with Applications*, vol. 109, pp. 1-11, <https://doi.org/10.1016/j.eswa.2018.05.011>.

Lahmiri, S & Bekiros, S 2019, 'Cryptocurrency forecasting with deep learning chaotic neural networks', *Chaos, Solitons & Fractals*, vol. 118, pp. 35-40, <https://doi.org/10.1016/j.chaos.2018.11.014>.

Manchanda, H & Aggarwal, S 2021, 'Forecasting cryptocurrency time series using AdaBoost-based ensemble learning techniques', in *Innovations in Cyber Physical Systems*, Springer, Singapore, pp. 207-219, https://doi.org/10.1007/978-981-16-4149-7_18.

McNally, S, Roche, J & Caton, S 2018, 'Predicting the price of Bitcoin using machine learning', in *2018 26th Euromicro International Conference on Parallel, Distributed and Network-Based Processing (PDP)*, IEEE, pp. 339-343, <https://doi.org/10.1109/PDP2018.2018.00060>.

Murray, K, Rossi, A, Carraro, D & Visentin, A 2023, 'On forecasting cryptocurrency prices: A comparison of machine learning, deep learning, and ensembles', *Forecasting*, vol. 5, no. 1, pp. 196-209, <https://doi.org/10.3390/forecast5010010>.

Nakano, M, Takahashi, A & Takahashi, S 2018, 'Bitcoin technical trading with artificial neural network', *Physica A: Statistical Mechanics and its Applications*, vol. 510, pp. 587-609, <https://doi.org/10.1016/j.physa.2018.07.017>.

Ortu, M, Uras, N, Conversano, C, Bartolucci, S & Destefanis, G 2022, 'On technical trading and social media indicators for cryptocurrency price classification through deep learning', *Expert Systems with Applications*, vol. 198, article 116804, <https://doi.org/10.1016/j.eswa.2022.116804>.

Rahmani Cherati, M, Haeri, A & Ghannadpour, SF 2021, 'Cryptocurrency direction forecasting using deep learning algorithms', *Journal of Statistical Computation and*

Simulation, vol. 91, no. 12, pp. 2475-2489,
<https://doi.org/10.1080/00949655.2021.1899179>.

Sebastião, H & Godinho, P 2021, 'Forecasting and trading cryptocurrencies with machine learning under changing market conditions', *Financial Innovation*, vol. 7, no. 1, article 3, <https://doi.org/10.1186/s40854-020-00217-x>.

Sun, X, Liu, M & Sima, Z 2020, 'A novel cryptocurrency price trend forecasting model based on LightGBM', *Finance Research Letters*, vol. 32, article 101084,
<https://doi.org/10.1016/j.frl.2018.12.032>.

Tuwiner, J 2022, *63+ cryptocurrency statistics, facts & trends*, BuyBitcoinWorldwide, viewed 14 May 2026, <https://buybitcoinworldwide.com/cryptocurrency-statistics/>.

Walther, T, Klein, T & Bouri, E 2019, 'Exogenous drivers of bitcoin and cryptocurrency volatility – A mixed data sampling approach to forecasting', *Journal of International Financial Markets, Institutions and Money*, vol. 63, article 101133,
<https://doi.org/10.1016/j.intfin.2019.101133>.

Zhang, J, Cai, K & Wen, J 2024, 'A survey of deep learning applications in cryptocurrency', *iScience*, vol. 27, no. 1, article 108509,
<https://doi.org/10.1016/j.isci.2023.108509>.

Zhang, W, Wang, P, Li, X & Shen, D 2018, 'Some stylized facts of the cryptocurrency market', *Applied Economics*, vol. 50, no. 55, pp. 5950-5965,
<https://doi.org/10.1080/00036846.2018.1488076>.

Zhang, Z, Dai, HN, Zhou, J, Mondal, SK, García, MM & Wang, H 2021, 'Forecasting cryptocurrency price using convolutional neural networks with weighted and attentive memory channels', *Expert Systems with Applications*, vol. 183, article 115378,
<https://doi.org/10.1016/j.eswa.2021.115378>.

Appendix

Appendix A: Data Dictionary

Feature	Data Type	Description
date	Date	Trading date of the observation.

open	Float	Opening price at the beginning of the trading day.
high	Float	Highest price during the trading day.
low	Float	Lowest price during the trading day.
close	Float	Closing price at the end of the trading day.
volume	Float	Total trading volume during the trading day.
quote_asset_volume	Float	Total traded value in the quote asset during the trading day.
number_of_trades	Integer	Total number of executed trades during the trading day.
taker_buy_base_asset_volume	Float	Volume of the base asset bought by takers during the trading day.
taker_buy_quote_asset_volume	Float	Value of the quote asset spent by takers buying during the trading day.
daily_return	Float	Percentage change in closing price compared with the previous day.
log_return	Float	Logarithmic return of the closing price relative to the previous day.
volatility_7	Float	Rolling standard deviation of returns over the past 7 days.
volatility_14	Float	Rolling standard deviation of returns over the past 14 days.
volatility_30	Float	Rolling standard deviation of returns over the past 30 days.
MA_7	Float	7-day simple moving average of the closing price.
MA_14	Float	14-day simple moving average of the closing price.
MA_30	Float	30-day simple moving average of the closing price.

EMA_7	Float	7-day exponential moving average of the closing price.
EMA_14	Float	14-day exponential moving average of the closing price.
EMA_30	Float	30-day exponential moving average of the closing price.
RSI_7	Float	7-day Relative Strength Index.
RSI_14	Float	14-day Relative Strength Index.
RSI_30	Float	30-day Relative Strength Index.
momentum_7	Float	Price momentum calculated over the past 7 days.
momentum_14	Float	Price momentum calculated over the past 14 days.
momentum_30	Float	Price momentum calculated over the past 30 days.
volume_MA_7	Float	7-day moving average of trading volume.
volume_MA_14	Float	14-day moving average of trading volume.
volume_MA_30	Float	30-day moving average of trading volume.